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SEEK-Policy: a data-driven cross-modal retrieval framework for climate policy recommendation

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ABSTRACT

Timely and context-sensitive climate policy recommendation requires intelligent systems that can seam-lessly integrate evolving environmental trends with domain-specific knowledge. However, existing tools rely predominantly on static keyword-based searches, lacking the ability to combine structured time-series signals with unstructured policy texts or to perform semantic reasoning across modalities. This limitation hinders policymakers from formulating rapid, evidence-based responses to emerging climate challenges, especially in resource-limited and data-constrained contexts. In this work, we present SEEK-Policy (Semantic Embeddings for Environment and Knowledge-based Policy) an open and reproducible cross-modal retrieval framework that directly aligns multivariate time-series data with climate policy documents using semantic embeddings and retrieval-augmented generation (RAG). The framework introduces three key modules: TimeTranscriber, which transforms multivariate time-series signals into concise, policy-relevant summaries; ChunkAlign, which performs fine-grained retrieval by mapping these summaries to semantically aligned segments within large corpora of climate policy documents; and PolicySynthesizer, which composes these segments into coherent, interpretable recommendations. To support open science, we develop POLiMATCH, a dataset linking over 1,200 climate policy documents with 10 years of structured time-series data across multiple countries and sectors, enabling rigorous evaluation of cross-modal retrieval tasks. Empirical results demonstrate that SEEK-Policy consistently outperforms strong baselines, including a Siamese network with contrastive learning and a role-based multi-agent summarization approach, achieving improvements of approximately 22% in Hit@1 and 8% in MRR@5 over the best-performing baseline. Beyond retrieval performance, SEEK-Policy provides an interpretable pipeline that links quantitative environmental indicators to climate policy documents, aiming to reduce the technical burden on decision-makers in resource-constrained settings. The framework and POLiMATCH dataset are openly available to support further research in cross-modal climate policy retrieval.

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Additional Information and Declarations can be found on page 39

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INTRODUCTION

Climate policy is a fundamental tool in addressing global challenges such as climate change, energy transitions, and sustainability. These policies help reduce emissions, protect ecosystems, and enhance economic and social resilience by providing a structured framework for governments and organizations. Effective climate governance increasingly depends on the ability to design policies that are data-driven, adaptive, and responsive to dynamic environmental conditions. Coordinated action across sectors ensures that global efforts to combat climate change remain aligned with long-term sustainability goals. The increasing frequency of extreme weather events and the need for global cooperation emphasize the urgency of evolving climate policies. Furthermore, the effectiveness of these policies can vary by region, highlighting the need for methods that integrate local socio-economic and environmental conditions into policy analysis (*Zheng et al., 2023*).

Policymakers require advanced tools to navigate the complexities of climate policy, particularly amid rapidly evolving data and international standards. Existing policy document recommendation systems, such as those provided by *Climate Policy Radar (2023)*, Climate Policy Database (CPDB) (*NewClimate Institute, Wageningen University & Research and PBL Netherlands Environmental Assessment Agency, 2023*), and Climate Change Laws of the World (CCLW) (*Grantham Research Institute at the London School of Economics and Climate Policy Radar, 2023*), typically rely on static, text-based queries where users input keywords or phrases to retrieve relevant documents. This reliance on static queries results in delayed and less effective policy responses, especially when urgent, time-sensitive actions are required. These systems cannot also integrate real-time data, limiting their effectiveness in guiding immediate decision-making (*Sakieh, 2023*; *Sietsma et al., 2024*). This challenge is especially significant in resource-limited and data-constrained contexts, where analytical capacity may be insufficient for large-scale policy evaluation.

Time-series data, such as temperature trends and emissions levels, are critical for shaping climate policies because they reflect ongoing environmental changes in real time. However, traditional text-based recommendation systems cannot seamlessly integrate this dynamic data. These systems typically rely on manual processes, requiring policymakers to analyze real-time data separately and cross-reference it with static documents. This labor-intensive approach delays policy formulation and hinders responsiveness, particularly when urgent, time-sensitive actions are needed (*Sewerin et al., 2023*; *Enenkel, Guha-Sapir & Zaitchik, 2024*). As a result, the accuracy and timeliness of policy recommendations are further compromised. Bridging this gap requires advanced computational methods that unify time-series analysis with semantic retrieval over large-scale policy corpora (*Poirier, Hermes & Aliberti, 2024*).

To overcome these limitations, we propose SEEK-Policy (Semantic Embeddings for Environment and Knowledge-based Policy), a cross-modal retrieval framework that aligns multivariate time-series data with climate policy documents. The system leverages large language models (LLMs) and retrieval-augmented generation (RAG) to provide interpretable recommendations that reduce the cognitive and technical burden on

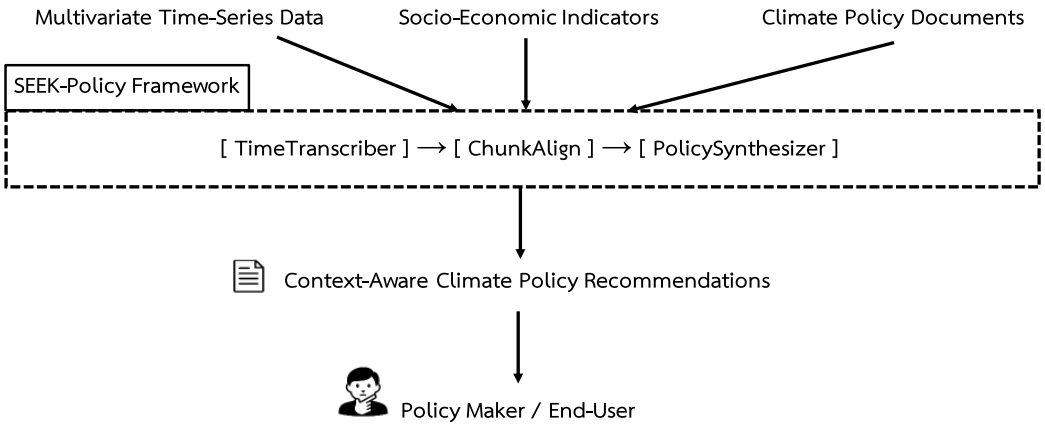


Figure 1 Overview of the SEEK-Policy framework. Existing systems rely on static keyword search and cannot link real-time data with policy texts. SEEK-Policy integrates time-series analysis and semantic retrieval through its TimeTranscriber, ChunkAlign, and PolicySynthesizer modules to generate context-aware climate policy recommendations. Full-size DOI: 10.7717/peerj-cs.4054/fig-1

decision-makers. SEEK-Policy introduces three key components: (1) TimeTranscriber, which generates policy-relevant natural language summaries from multivariate time-series data; (2) ChunkAlign, which retrieves semantically aligned text segments from policy corpora using embedding-based retrieval; and (3) PolicySynthesizer, which composes retrieved segments into coherent and interpretable recommendations. An overview of this framework is shown in Fig. 1.

To support reproducibility and open science, we also present POLiMATCH, a new dataset linking over 1,200 climate policy documents with 10 years of historical time-series data across multiple countries and sectors. This dataset enables rigorous benchmarking of cross-modal retrieval tasks and contributes an open resource for the research community.

We compare SEEK-Policy against two strong baselines: a contrastive learning approach using a Siamese network to embed time series and policy summaries into a shared latent space (*Jing et al., 2024*; *Dong et al., 2024*), and a multi-agent LLM-based summarization framework that simulates domain-specific policy advisors (*Xiao et al., 2024*; *Han et al., 2025*). Experimental results show that SEEK-Policy consistently outperforms all automatic baselines across multiple retrieval metrics. In contrast, a human-oracle baseline, where expert-written summaries serve as retrieval queries, establishes an upper bound that reflects the ceiling achievable with perfect policy understanding. The remaining gap between automated and human performance opens a promising research direction in cross-modal query generation from quantitative environmental data, which remains largely unexplored in the climate policy retrieval literature.

The key contributions of this article are summarized below:

- We identify a critical gap in current policy support systems: the lack of methods that integrate multivariate time-series data with unstructured policy texts in a context-aware and semantically meaningful way.

- We propose SEEK-Policy as a systems-level cross-modal retrieval and policy recommendation framework that integrates time-series-to-text transformation, semantic chunk retrieval, policy-oriented synthesis, and reproducible evaluation.
- We introduce three core modules: *TimeTranscriber*, *ChunkAlign*, and *PolicySynthesizer*, which transform structured time-series into policy-relevant summaries, retrieve semantically aligned text segments, and synthesize interpretable recommendations.
- We develop POLiMATCH, a new open dataset that links more than 1,200 climate policy documents with 10 years of time-series data across countries and sectors.
- We benchmark SEEK-Policy against state-of-the-art baselines and demonstrate consistent improvements in retrieval accuracy, particularly in unfiltered scenarios requiring contextual reasoning.
- We highlight the framework's accessibility and potential impact in resource-limited and data-constrained contexts, ensuring broader applicability beyond high-capacity research environments.

The remainder of this article is structured as follows: 'Related Work' reviews related work in semantic retrieval, multimodal AI, and policy recommendation. 'Preliminaries and Problem Definition' and 'The Proposed Framework: SEEK-Policy' defines the problem formally. 'Experimental Setting' describes the experimental setup and the SEEK-Policy architecture. 'Experimental Results' presents evaluation results. The 'Discussion' discusses implications for climate policy support, and 'Conclusion and Future Work' concludes with future research directions.

RELATED WORK

Climate policy recommendation systems have emerged as essential tools for navigating the growing complexity of environmental governance. Platforms such as *Climate Policy Radar (2023)*, the Climate Policy Database (CPDB) (*NewClimate Institute, Wageningen University & Research and PBL Netherlands Environmental Assessment Agency, 2023*), and Climate Change Laws of the World (CCLW) (*Grantham Research Institute at the London School of Economics and Climate Policy Radar, 2023*) offer structured access to climate legislation and national commitments across countries and sectors. CPDB, in particular, has been maintained by the NewClimate Institute under the EU H2020 initiative since 2016 and is widely used for comparative policy analysis. However, these platforms primarily function as search engines based on static keyword queries and lack mechanisms to integrate structured environmental or socio-economic data with unstructured text. As a result, their utility in time-sensitive decision-making and context-specific policy formulation remains limited.

While there is increasing interest in data-driven climate policy, most systems stop at document retrieval rather than providing actionable recommendations. This disconnect between static information access and the evolving nature of climate challenges underscores the need for adaptive, real-time systems that can align policies with dynamic time-series signals (*Sakieh, 2023*; *Sietsma et al., 2024*).

Contrastive learning for policy and document retrieval

Contrastive learning techniques have gained popularity for document matching tasks due to their ability to encode semantically meaningful relationships. Siamese networks, in particular, have been widely applied to sentence and document embeddings using contrastive or triplet loss (*Reimers & Gurevych, 2019a*; *Gao, Yao & Chen, 2021*). These models embed both queries and documents into a shared latent space and retrieve relevant results based on vector similarity. Such approaches have been adopted in legal and policy domains to enhance the precision of document retrieval systems. In our framework, we implement a contrastive learning baseline that embeds both time-series signals and sectoral summaries into a shared space, enabling direct comparison against SEEK-Policy's cross-modal retrieval pipeline.

Multi-agent and role-based systems for document generation

Multi-agent prompting and role-specific language modeling have emerged as strategies for simulating expert reasoning across domains such as healthcare, law, and governance. These systems leverage role definitions—such as economic advisors or climate analysts—to guide large language models (LLMs) in generating more structured and perspective-aware outputs. For example, *Park et al. (2023)* used autonomous generative agents in simulation environments. In contrast, *Spokoyny et al. (2024)* employed role-based prompting to align policy content with UN Sustainable Development Goals. Our baseline adopts a similar multi-agent design in which different roles interpret time-series data and generate corresponding policy-oriented summaries, providing a point of comparison for SEEK-Policy's integrated semantic retrieval and synthesis approach.

NLP, semantic search, and retrieval-augmented generation in climate policy

Natural Language Processing (NLP) methods have been widely adopted in climate-related research and policy analysis. Tasks include identifying misinformation (*Farrell, 2019*), extracting sustainability metrics from text (*Smith et al., 2023*), and analyzing climate-related financial disclosures (*Luccioni, Baylor & Duchene, 2020*). Domain-specific models such as ClimateBERT (*Webersinke et al., 2022*) have demonstrated improved performance on classification and retrieval tasks by tailoring transformer architectures to the language of environmental policy.

Semantic search has further advanced the utility of NLP in this domain. Sentence-BERT (SBERT) (*Reimers & Gurevych, 2019b*), and similar models offer embedding-based retrieval that moves beyond surface keyword matching to capture deeper contextual meaning. This is especially beneficial in legal and policy datasets, which often contain complex terminology and hierarchical structure (*Roelfsema et al., 2020*; *Spokoyny et al., 2024*). Retrieval-Augmented Generation (RAG) builds on these advances by fusing information retrieval with generative modeling. RAG-based systems have been used successfully in domains such as biomedical literature search and clinical diagnostics (*Siriwardhana et al., 2023*; *Munarko, Rampadarath & Nickerson, 2023*; *Yu, Guo & Sano, 2023*), enabling LLMs to ground their outputs in evidence-based sources. While

general-purpose LLMs like GPT-4 offer powerful generative capabilities, they are limited in handling structured or temporal data without retrieval mechanisms (*Cadamuro et al., 2023*). In SEEK-Policy, we introduce a two-stage framework that leverages this paradigm: *TimeTranscriber* converts time-series inputs into concise policy-oriented summaries, and *ChunkAlign* retrieves aligned document passages using semantic vector search, providing interpretable recommendations that bridge dynamic data with legally grounded climate policy text.

Time-series integration in policy AI systems

Time-series data, such as greenhouse gas emissions, temperature anomalies, and energy use, are crucial for understanding climate trends and crafting timely policies. *Ferguson et al. (2012)* demonstrated the impact of long-term ecological datasets in guiding conservation decisions in the Arctic. HYBTEP (*Fortes et al., 2014*), a hybrid modeling framework, used macroeconomic indicators to simulate the consequences of various climate policies. TCN-LSTM models have recently been deployed for CO_2 forecasting, improving the alignment between emission predictions and mitigation strategies (*Wei & Xu, 2023*).

However, substantial challenges remain in integrating this structured, high-frequency data with unstructured textual policy documents. *Thorne et al. (2017)* highlighted the fragmentation of sectoral data and its impact on cross-sectoral policy design. *Poirier, Hermes & Aliberti (2024)* noted persistent data gaps in developing regions, limiting the global consistency of policy evaluation. *Jin et al. (2024b)* and *Cheng & Chin (2024)* recently proposed frameworks that begin to integrate time-series data with LLMs. However, they remain restricted in scalability, interpretability, and generalization across diverse domains.

Summary of key differences

While prior work has separately explored contrastive learning, multi-agent prompting, semantic search, and time-series integration, our model is the first to unify these components into a full-stack, cross-modal system for climate policy recommendation. SEEK-Policy introduces a novel two-stage pipeline: TimeTranscriber converts dynamic, multivariate time-series data into natural-language summaries, and *ChunkAlign* retrieves semantically aligned document segments *via* Retrieval-Augmented Generation. Together with the PolicySynthesizer and the POLiMATCH dataset, this framework advances beyond prior art by delivering scalable, interpretable, and real-time recommendations that integrate structured and unstructured data for policy decision support.

PRELIMINARIES AND PROBLEM DEFINITION

We begin by introducing the basic notation and formal definitions used throughout this work. SEEK-Policy is designed to align multivariate time-series signals with climate policy documents through semantic transformation and retrieval.

Definition 1 (Time-series input). Let $X = \{x_1, x_2, \ldots, x_T\}$, where each $x_t \in \mathbb{R}^d$ denotes the observation at time step t across d-dimensional indicators such as greenhouse gas emissions, energy use, or GDP. Each input sequence is associated with a sectoral context

$S' \subseteq S$, where S is the set of all possible sectors, and a region identifier R representing the country or geographic unit of analysis.

Definition 2 (Policy document pool). Let $P = \{p_1, p_2, \ldots, p_N\}$ denote the *corpus* of climate policy documents. Each document p_j is segmented into a sequence of coherent textual units (chunks):

$$p_j = \{c_j^1, c_j^2, \ldots, c_j^{m_j}\},$$

where m_j is the number of chunks in document p_j. The union of all such chunks across the *corpus* defines the retrieval pool:

$$C = \bigcup_{j=1}^{N} \{c_j^1, \ldots, c_j^{m_j}\}.$$

Definition 3 (Semantic transformation and embedding functions). We define the following functions that form the basis of the SEEK-Policy framework:

- $f_{\text{trans}}(X, S')$: a transformation function that generates a semantically rich natural language query q from the time-series sequence X and its sectoral context S'. This is implemented *via* the *TimeTranscriber* module.
- $f_{\text{embed}}(q)$: an embedding function that maps the query q into a semantic vector space.
- $g(c)$: an embedding function that maps each policy chunk $c \in C$ into the same vector space for similarity-based retrieval.

Problem definition (SEEK-Policy). Given a multivariate time-series input X, sectoral context S', and policy document *corpus* P, the objective of SEEK-Policy is to identify a set of top-N documents $\hat{P} \subseteq P$ that are most semantically aligned with the evolving environmental and governance context.

This is achieved in three stages: (1) generating a semantically meaningful query q from the input time-series data and sectoral metadata; (2) retrieving policy chunks $c \in C$ that are closest to q in the shared embedding space; and (3) aggregating these chunks to rank and return the most relevant documents.

THE PROPOSED FRAMEWORK: SEEK-POLICY

The SEEK-Policy framework is a modular pipeline designed to align multivariate time-series data with semantically relevant climate policy documents. It comprises three main components: the TimeTranscriber, which transforms multivariate indicators into natural language summaries; the ChunkAlign, which retrieves semantically aligned text passages from policy corpora; and the PolicySynthesizer, which composes the retrieved chunks into unified, interpretable outputs. The framework is trained and evaluated using the POLiMATCH dataset, a curated *corpus* that links time-series data with sector-specific policy documents across countries and regions. An overview of SEEK-Policy framework is shown in Fig. 2.

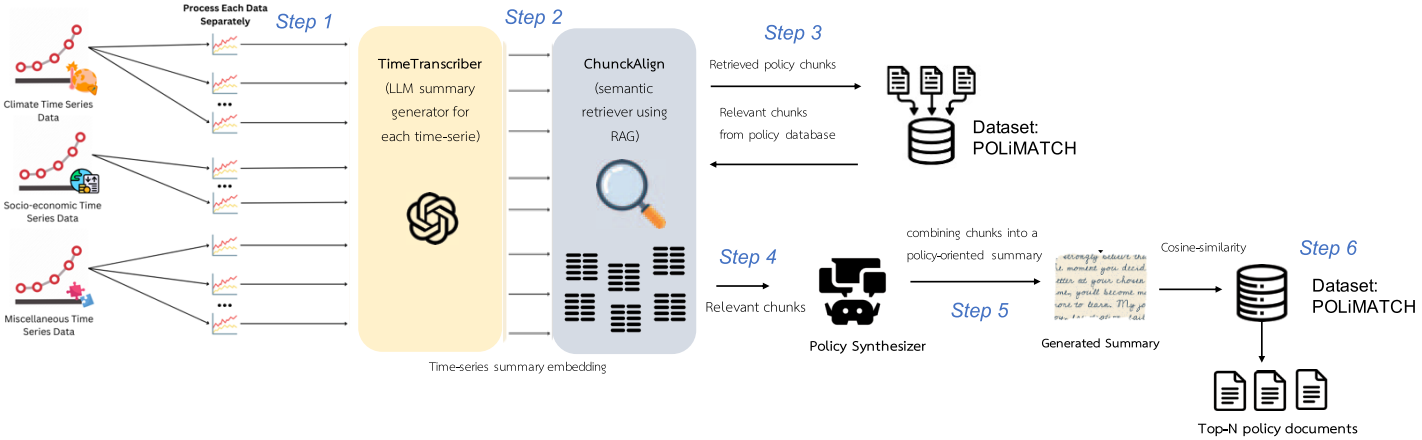


Figure 2 Architecture of the proposed SEEK-Policy framework. The system ingests multivariate time-series data from climate, socio-economic, and environmental sources. Each sequence is summarized using the TimeTranscriber module, which generates a semantic natural language query. The query is passed to the ChunkAlign module, a semantic retriever based on Retrieval-Augmented Generation (RAG), which identifies the most relevant policy chunks from the POLiMATCH dataset. These chunks are synthesized into a final, interpretable recommendation by the Policy-Synthesizer module, producing context-aware climate policy insights. Full-size DOI: 10.7717/peerj-cs.4054/fig-2

Time-series selection and summary generation (TimeTranscriber)

The pipeline begins by selecting the most relevant time-series tables for a given policy task. Each policy document in POLiMATCH is associated with metadata, including targeted sector, policy instrument, keywords, topics, and climate hazards. A prompt-based large language model (LLM) procedure, referred to as the Prompt for Selecting Relevant Time-Series Data, filters the available tables using this metadata.

```
Based on this table's names: {table_names}, select all that are directly related to the policy with the following metadata.
Sector: {sector}
Instrument: {instrument}
Keywords: {keywords}
Topics: {topics}
Hazards: {hazards}
Return as a list with no explanation.
```

The selected tables are summarized into sector-specific paragraphs *via* the TimeTranscriber module, implemented with an LLM:

```
You are a policy analyzer. Analyze the following data, focusing on the sector {sector}.
Title: {table_title}
Description: {dataset description}
Country Data: {country_data}
Region Data: {region_data}
```

World Data: {world_data}
Output: one paragraph only.

The output is a concise, semantically rich summary q that serves as the query for semantic retrieval.

Semantic chunk retrieval (ChunkAlign)

Given the query q, the ChunkAlign module retrieves relevant policy content from the POLiMATCH *corpus*. Each policy document is pre-segmented into coherent text chunks $c \in C$, where C is the retrieval pool. Both the summary query and each chunk are encoded into a shared semantic embedding space:

$$v_q = f_{\text{embed}}(q), \quad v_c = g(c), \quad c \in C$$

Semantic similarity is computed using cosine similarity:

$$\text{Sim}(v_q, v_c) = \frac{v_q \cdot v_c}{\|v_q\|\|v_c\|}$$

where $v_q = f_{\text{embed}}(q)$ and $v_c = g(c)$ are the respective embeddings. The top-k most similar chunks form the set $C_{\text{retrieved}} \subseteq C$, which are passed to the next stage for synthesis. This chunk-level retrieval approach, rather than whole-document comparison, enables precise alignment with specific parts of policy text that reflect the intent and structure of the query, grounded in real-time environmental signals.

The top-k most similar chunks form the set $C_{\text{retrieved}} \subseteq C$, which are passed to the synthesis stage. This chunk-level retrieval ensures fine-grained alignment with policy text, enabling higher precision compared to whole-document retrieval.

Policy-oriented summary composition (PolicySynthesizer)

The PolicySynthesizer module aggregates retrieved chunks into a coherent, interpretable summary that reflects both the original time-series input and the corresponding policy context. The following prompt guides the LLM-based synthesis:

You are a policy analyzer. Summarize the following chunks into one paragraph.
Chunks: {CHUNK_1}, {CHUNK_2},…, {CHUNK_n}
Output: one paragraph only.

Instead of retrieving entire documents, SEEK-Policy retrieves and synthesizes semantically aligned text segments, enabling granular understanding of relevant policy measures. Retrieved chunks are traced back to their source documents, which are then ranked based on aggregated similarity scores or frequency of retrieved segments. The top-N policy documents are then returned as recommendations alongside the synthesized summary, enhancing transparency and interpretability.

Algorithm 1 **SEEK-Policy: data-driven climate policy recommendation.**

Input: Time-series data X; sector and region metadata (S', R); policy document set P with chunk pool C.

Output: Synthesized policy summary $\hat{y}$; top-N policy documents $\hat{P}$.

Step 1: Time-Series Table Selection
$T_{\text{selected}} \leftarrow \text{SelectTables}(X, S', R)$ using LLM and metadata.
Step 2: Semantic Summary Generation (TimeTranscriber)
$q \leftarrow \text{TimeTranscriber}(T_{\text{selected}})$.
Step 3: Chunk-Level Semantic Retrieval (ChunkAlign)
foreach $c \in C$ **do**
    Compute similarity $\text{Sim}(q, c) = \frac{v_q \cdot v_c}{\|v_q\| \|v_c\|}$.
$C_{\text{retrieved}} \leftarrow \text{Top}-k$ chunks.
Step 4: Chunk Summarization (PolicySynthesizer)
$\hat{y} \leftarrow \text{PolicySynthesizer}(C_{\text{retrieved}})$.
Step 5: Document Ranking and Filtering
Rank documents by the aggregated similarity of retrieved chunks. Apply sector/country filters and deduplication. $\hat{P} \leftarrow \text{Top}-N$ documents.
return $\hat{y}, \hat{P}$.

EXPERIMENTAL SETTING

Dataset: POLiMATCH

To support context-aware climate policy recommendation, we constructed the POLiMATCH dataset (Policy and Linked Multimodal Alignment of Time-Series and Climate Histories). POLiMATCH integrates structured environmental indicators with annotated climate policy documents, enabling training and evaluation of models that align real-world trends with policy content at both the document and chunk levels[1].

[1] The code, dataset, preprocessing scripts, evaluation artifacts, embeddings, and reproduction instructions are available at: https://github.com/thamolwanpo/SEEK-Policy.

Data collection and sources

Policy documents were collected from Climate Policy Radar (*Climate Policy Radar, 2023*), an open-access repository comprising more than 7,000 climate-related laws, policies, and UNFCCC submissions. Each document is accompanied by metadata such as title, summary, sector(s), policy instrument, country, enactment date, and timeline events. After filtering for metadata completeness and accessibility, 4,534 documents were retained for analysis. Of these, 1,243 contained English-language summaries and were used for retrieval and evaluation.

Time-series data were sourced from Our World In Data (OWID) (*Our World in Data, 2024*), encompassing 127 curated tables categorized as: 34 climate-related datasets, 50 socio-economic datasets, and 43 datasets on related topics (*e.g.*, renewable energy, waste, agriculture).

Figure 3 illustrates the distribution of tables across subgroups aligned with sectors targeted by our models. This categorization facilitates linking time-series inputs to sector-specific policy documents for accurate, context-aware recommendations.

Together, these tables cover more than 13,200 indicators across 120 global topics and provide annual data for over 200 countries.

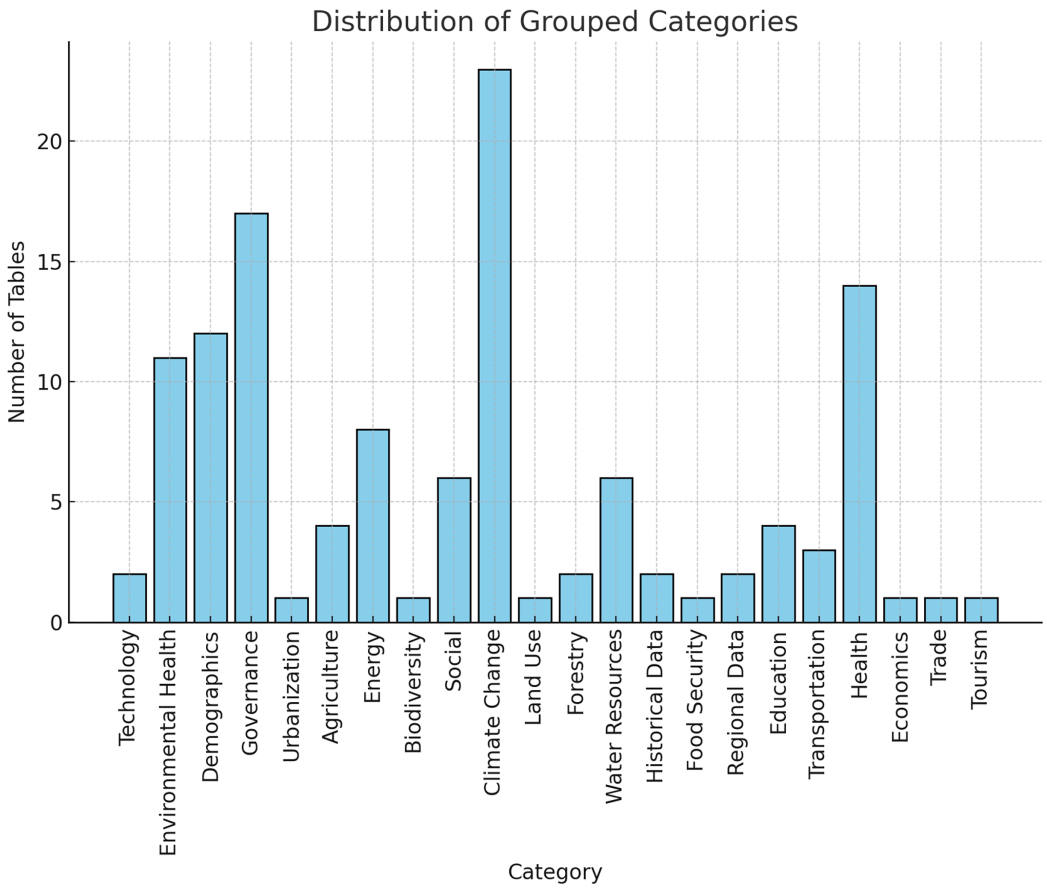


Figure 3 Distribution of time-series tables from Our World In Data across major categories, including climate change, health, demographics, and energy.
Full-size DOI: 10.7717/peerj-cs.4054/fig-3

Preprocessing and data alignment

Policy preprocessing. Documents lacking valid sector labels or timeline metadata were excluded. HTML tags were removed from summaries, and country identifiers were standardized using ISO alpha-3 codes.

Time-series preprocessing. OWID tables underwent a multi-stage pipeline to ensure consistency, completeness, and suitability for modeling. Each dataset was first aligned by `country` and `year` to standardize temporal and geographic resolution. We then filtered columns using three criteria: variables with missing values exceeding 20% of observations were removed, columns that were constant across years were discarded, and variables exhibiting pairwise correlations above 0.7 were excluded to reduce redundancy and multicollinearity. Importantly, these thresholds were applied exclusively to training data to prevent leakage into the test set. Remaining gaps were imputed in two stages: linear interpolation to fill internal breaks in trajectories, followed by backfilling to address boundary cases. To improve storage and computational efficiency, numeric columns were

downcast where appropriate (*e.g.*, `float64` to `float32` with negligible precision loss). Finally, country-level time series were composed by merging the preprocessed datasets by year; overlapping variable names were suffixed to preserve distinctions and avoid collisions. The resulting tables provide a multivariate profile for each country, temporally aligned with policy timelines.

Matching policy data and time-series data. To align real-world trends with policy actions, each policy document was paired with a corresponding time-series segment based on its country of origin and the last recorded policy event. To examine the effect of temporal context on retrieval performance, we constructed historical windows of $w \in \{1, 2, 5, 10\}$ years, each ending at the final event year in the policy's metadata timeline, rather than fixing a single arbitrary horizon. When fewer than w years of observations were available, zero-padding was applied to the leading time steps to preserve dimensional consistency across samples. Zero-padding was chosen over alternatives such as mean imputation or sample exclusion because it makes the absence of data explicit to the model rather than introducing potentially misleading values, and because excluding short-history countries would disproportionately remove developing nations with limited historical records, introducing geographic bias into the *corpus*. Importantly, the impact of zero-padding is empirically assessed through our multi-window evaluation: since $w = 10$ requires the most padding by construction, any degradation attributable to padding would manifest as lower retrieval performance at this horizon. The fact that $w = 10$ achieves the best retrieval performance across all configurations (Experimental Results) provides empirical evidence that zero-padding does not adversely affect model performance.

Feature representation

Sector encoding. Each policy may target one or more sectors (*e.g.*, energy, transport, agriculture). Sectoral metadata are encoded as a multi-hot vector $\mathbf{v}_P \in \{0, 1\}^{|S|}$, where $|S|$ is the number of predefined sectors:

$$\mathbf{v}_P[i] = \begin{cases} 1 & \text{if } s_i \in S_P, \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

where s_i denotes the i-th sector in the full set S, and $S_P \subseteq S$ is the sector subset associated with policy P. Policies that span multiple sectors have multiple entries set to 1, capturing their full scope.

Feature normalization. All numeric time-series features were standardized using z-score normalization:

$$X_i' = \frac{X_i - \mu_i}{\sigma_i}, \tag{2}$$

where μ_i and σ_i denote the mean and standard deviation of feature X_i.

Training and test set preparation

To evaluate the SEEK-policy, we focused on English-language policy documents with available summaries ($n = 1{,}243$). To ensure the statistical robustness of our retrieval

Table 1 POLiMATCH dataset statistics.

Data subset	# Documents
All retrieved policy documents	4,534
Total English policy documents	1,243
Evaluation protocol	5-fold group K-fold
Avg. test samples per fold	248
Total evaluation queries (aggregate)	1,243

evaluation and prevent data leakage, we implemented a group K-Fold cross-validation strategy. The dataset statistics is as shown in Table 1.

Leakage prevention (policy families): We identified and grouped policy families, which are documents representing different versions, amendments, or updates of the same primary legislation (*e.g.*, the UK Climate Change Act 2008 and its 2018 amendment). To prevent the model from seeing future or past versions of a test document during training, we ensured that all documents belonging to the same policy family were assigned to the same fold. This ensures a strict blind evaluation where the model cannot rely on overlapping legal text from the same legislative lineage.

Cross-validation and query generation: Given the limited size of the annotated *corpus* (1,243 samples), a static train/test split would yield a test set too small for statistically significant retrieval metrics. Instead, we employed 5-fold Group K-Fold cross-validation. In each fold, approximately 250 documents were held out for testing. By aggregating results across all five folds, we generated a total of 1,243 test queries, which allows for more stable confidence intervals in our reporting of Hit@k and MRR.

Test set selection criteria. A stratified sampling procedure was used to ensure balance across regions and sectors, following these principles:

1. *Geographic representation*: at least one document from each major global region;
2. *Country diversity*: avoidance of duplicates from the same ISO alpha-3 country;
3. *Sectoral balance*: manual checks to verify broad sector distribution.

Dataset distribution. To assess generalizability and fairness, we analyzed the distribution across regions, countries, and sectors for each cross-validation fold. Europe and Asia are most represented across all folds, while Africa and Oceania remain underrepresented relative to their real-world policy activity. Energy is the most prevalent sector, followed by Transport, Economy-wide, and Buildings, with other domains (*e.g.*, Agriculture, Industry, LULUCF) also well represented. Full details of these distributions, including figures, are provided in 'Dataset Distribution across Cross-Validation Folds'.

Baseline

We compare SEEK-Policy against two strong baselines: a Siamese network model (*Reimers & Gurevych, 2019b*) trained with triplet loss to learn joint embeddings, and a role-based

multi-agent system (*Wang et al., 2023*) that generates summaries using domain-specific prompting.

Siamese network model

Overview. The dual-encoder network aligns time-series inputs with policy documents in a shared latent space and enables similarity-based retrieval conditioned on time-series and sectoral information. The pipeline comprises a training phase with either a triplet or a contrastive loss, followed by an inference phase using cosine similarity. Figure 4 depicts (Fig. 4A) the training phase and (Fig. 4B) the inference process.

Model architecture. The network contains two parallel branches. The first branch encodes policy summaries p_j using a pretrained transformer backbone (*e.g.*, ClimateBERT (*Webersinke et al., 2022*) or SBERT (*Reimers & Gurevych, 2019a*)). The pooled representation is projected into the shared embedding space:

$$\mathbf{v}_{\text{text}} = \mathbf{W}_{\text{text}}\,\text{Backbone}_{\text{pooled}}(p_j) + \mathbf{b}_{\text{text}}. \tag{3}$$

The second branch processes the flattened time series $\mathbf{x} \in \mathbb{R}^{T\cdot d}$ and the one-hot sector vector $S' \in \mathbb{R}^k$. The time series is encoded through a two-layer MLP with an intermediate hidden layer of size h:

$$\mathbf{v}_{\text{ts}} = \mathbf{W}_2\,\text{Dropout}(\text{ReLU}(\mathbf{W}_1\,\mathbf{x} + \mathbf{b}_1)) + \mathbf{b}_2. \tag{4}$$

The sector vector is projected into a low-dimensional latent sector representation:

$$\mathbf{v}_{S'} = \mathbf{W}_{S'}S' + \mathbf{b}_{S'}. \tag{5}$$

The time-series and sector embeddings are concatenated and projected into the final joint space of dimension d_e:

$$\mathbf{v}_{\text{ts+sector}} = \mathbf{W}_{\text{final}}[\mathbf{v}_{\text{ts}}; \mathbf{v}_{S'}] + \mathbf{b}_{\text{final}}. \tag{6}$$

Training with triplet and contrastive losses. Each training triplet uses the policy summary as the anchor $\mathbf{v}_A = \mathbf{v}_{\text{text}}$, a matched time-series-sector pair as the positive $\mathbf{v}_{\text{Pos}} = \mathbf{v}_{\text{ts+sector}}$, and a non-matching time-series-sector pair as the negative $\mathbf{v}_{\text{Neg}}$.

Under triplet loss, the model is trained to bring matched pairs closer than mismatched pairs:

$$L_{\text{triplet}} = \max\big(0, \|\mathbf{v}_A - \mathbf{v}_{\text{Pos}}\| - \|\mathbf{v}_A - \mathbf{v}_{\text{Neg}}\| + \alpha\big), \tag{7}$$

where α is a margin enforcing separation between positive and negative pairs.

Under contrastive loss (InfoNCE (*van den Oord, Li & Vinyals, 2018*)), in-batch negatives are used with a temperature parameter τ:

$$L_{\text{contrastive}} = \frac{1}{2}\left(\mathscr{L}(\mathbf{V}_{\text{ts}}, \mathbf{V}_{\text{text}}) + \mathscr{L}(\mathbf{V}_{\text{text}}, \mathbf{V}_{\text{ts}})\right), \tag{8}$$

where $\mathscr{L}(\mathbf{A}, \mathbf{B}) = \text{CrossEntropy}\big(\mathbf{A}_{\text{norm}}\mathbf{B}_{\text{norm}}^{\top}/\tau,\ \mathbf{I}\big)$ and $\mathbf{I}$ indexes matched pairs along the diagonal.

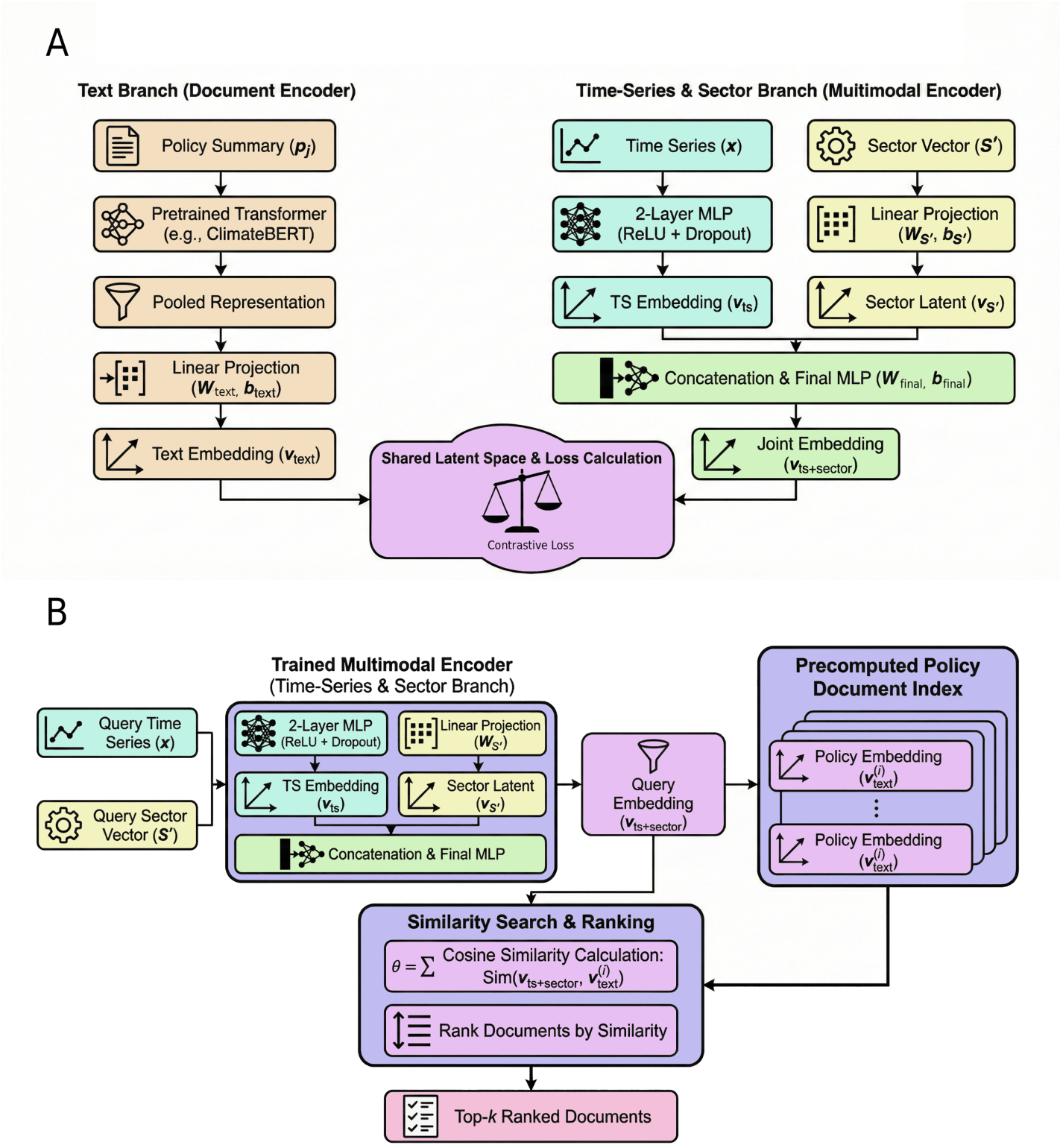


Figure 4 Siamese network baseline for climate policy retrieval. Dual encoders project time-series and textual inputs into a shared latent space for similarity-based recommendation. (A) Training with triplet loss. (B) Inference with nearest-neighbor search over precomputed policy vectors.
Full-size DOI: 10.7717/peerj-cs.4054/fig-4

Crucially, the entire architecture, including the pretrained transformer backbone, is fine-tuned end-to-end. All parameters are jointly optimized using AdamW with learning rate 1×10^{-4}, dropout, weight decay, and a `ReduceLROnPlateau` scheduler with early stopping.

Inference and semantic search. At inference, the trained encoder produces $\mathbf{v}_{\text{ts+sector}}$ from the input time series and sector metadata. Policy embeddings are precomputed with the text branch and projected into the shared space. Retrieval uses cosine similarity:

$$\text{Sim}(\mathbf{v}_{\text{ts+sector}}, \mathbf{v}_{\text{text}}) = \frac{\mathbf{v}_{\text{ts+sector}} \cdot \mathbf{v}_{\text{text}}}{\|\mathbf{v}_{\text{ts+sector}}\| \, \|\mathbf{v}_{\text{text}}\|}. \tag{9}$$

Documents are ranked by similarity, and the top-k results are returned. Figure 4 illustrates the inference pipeline.

Role-based multi-agent system

This baseline adopts a zero-shot, role-based multi-agent approach in which domain-specialized agents independently analyze different aspects of the input time series. Each agent acts as a policy advisor with expertise in a particular area (*e.g.*, climate policy, socio-economic trends, sustainable development) and receives sector-relevant time-series data curated from 'Dataset: POLiMATCH'. Agents are implemented *via* role-conditioned prompts that produce focused summaries without fine-tuning, following the role-playing paradigm of *Tseng et al. (2024)*. The advisors' outputs are passed to an interdisciplinary agent that integrates cross-domain insights, and a policy-summarization agent refines the combined text into a concise, coherent summary for decision-makers. This pipeline mirrors expert committee workflows, where multiple perspectives contribute to well-rounded recommendations. The architecture is shown in Fig. 5, and full prompt templates are provided in 'Prompts for Role-Based Agents Approach'.

Post-processing

All retrieval methods incorporate a document-level de-duplication step to ensure that each unique policy document appears at most once in the final recommendation set. Concretely, when multiple retrieved chunks originate from the same source document, only the chunk with the highest similarity score is retained as a representative of that document; all remaining chunks from the same document are discarded. The system then ranks all representative documents by their retained scores and returns the top-k unique policy documents. By construction, this procedure guarantees that no single source document can occupy more than one position in the output, fully preventing over-representation of any individual document in the recommendation set.

Evaluation metrics

We evaluate retrieval quality using three complementary metrics: Hit@k, Normalized Discounted Cumulative Gain (NDCG@k), and Mean Reciprocal Rank (MRR). Together, these metrics assess not only whether a relevant policy is retrieved, but also whether the

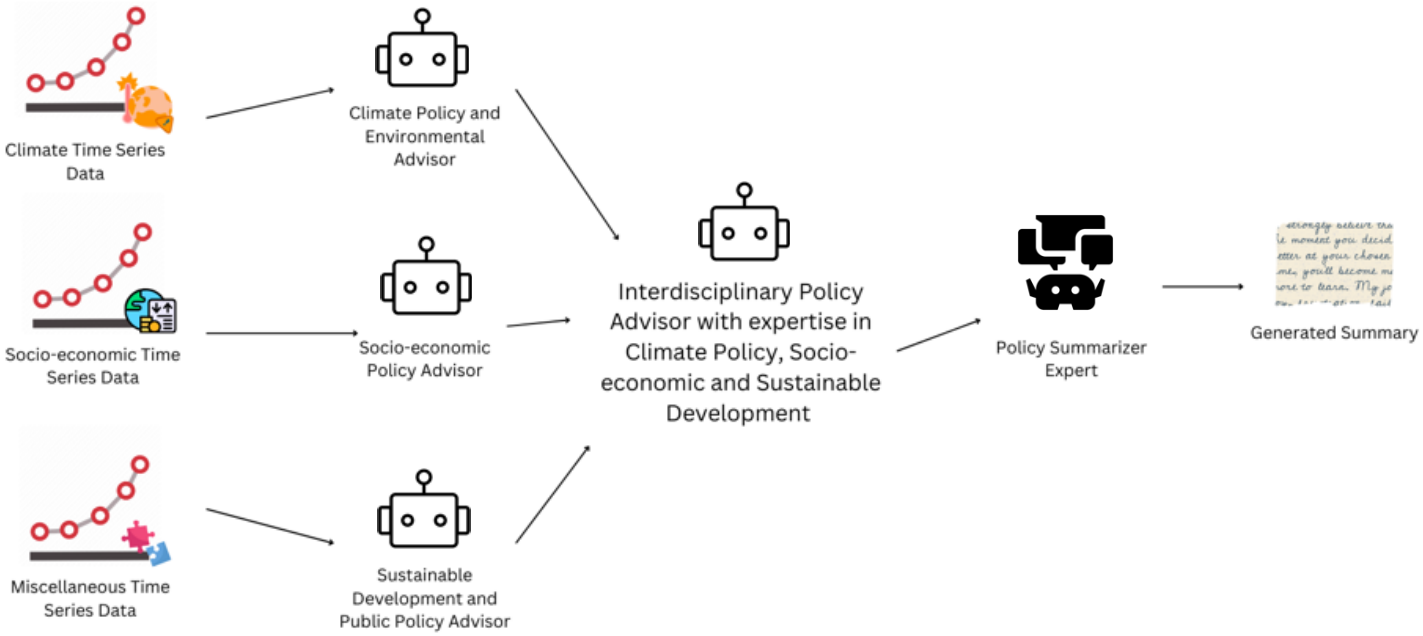


Figure 5 Architecture of the role-based multi-agent baseline. Domain-specific advisors analyze time-series inputs, an interdisciplinary agent integrates the content, and a policy-summarization agent produces the final output. Full-size DOI: 10.7717/peerj-cs.4054/fig-5

most actionable content is ranked highest, a critical requirement in climate policy decision-making, where top-ranked results must include specific policy instruments rather than general preamble statements.

Hit@k. Hit@k measures the fraction of queries for which a relevant result appears within the top k retrieved documents:

$$\text{Hit@k} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\{\text{rank}(r_i) \leq k\}, \tag{10}$$

where N is the number of queries, r_i is the rank of the relevant document for query i, and $\mathbb{1}\{\cdot\}$ is the indicator function. We report Hit@1, corresponding to exact top-1 correctness. While Hit@k confirms that relevant documents have been retrieved, it does not indicate whether the highest-ranked result contains actionable policy clauses or only introductory text.

Normalized Discounted Cumulative Gain (NDCG@k). To address this limitation, we additionally report NDCG@k, which rewards systems that place the most relevant content at higher ranks by applying a logarithmic discount to lower-ranked results:

$$\text{NDCG@k} = \frac{\text{DCG@k}}{\text{IDCG@k}}, \quad \text{where} \quad \text{DCG@k} = \sum_{j=1}^{k} \frac{\text{rel}_j}{\log_2(j+1)}, \tag{11}$$

where rel_j is the relevance score of the document at rank j, and IDCG@k is the ideal DCG achieved by a perfect ranking. We report NDCG@5 and NDCG@10.

Table 2 Retrieval performance at $w = 10$. Human summary (gold) serves as an expert upper bound and is not a competing system. Best automatic result in bold.

Method	Hit@1	NDCG@5	NDCG@10	MRR@5
Human summary (gold)	0.688	0.820	0.828	0.785
ClimatePolicyRadar (keyword) (*Climate Policy Radar, 2023*)	0.052	0.099	0.124	0.083
Siamese (ClimateBERT + contrastive)	0.304	0.496	0.533	0.438
Siamese (ClimateBERT + triplet)	0.053	0.134	0.169	0.108
Siamese (all-distilroberta + contrastive)	0.304	0.497	0.531	0.440
Siamese (all-distilroberta + triplet)	0.083	0.178	0.220	0.147
Role-Agent	0.301	0.431	0.449	0.395
SEEK-Policy	**0.370**	**0.515**	**0.535**	**0.474**

In the context of climate policy retrieval, a high NDCG indicates that passages containing specific, actionable policy instruments appear near the top of the ranked list, rather than being buried behind general or introductory content. This makes NDCG a more stringent and externally valid indicator of decision-support quality than Hit@k alone.

Mean Reciprocal Rank (MRR). MRR captures the average inverse rank of the first relevant result across queries:

$$\mathrm{MRR} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\mathrm{rank}(r_i)}, \tag{12}$$

where $\mathrm{rank}(r_i)$ is the position of the first relevant document for query i. MRR complements NDCG by emphasising how quickly the system surfaces the first actionable result, which is particularly important in time-sensitive policy recommendation scenarios. Higher values across all three metrics indicate stronger retrieval performance.

EXPERIMENTAL RESULTS

Overall performance

Table 2 presents retrieval performance across all methods using a historical window of $w = 10$ years. We include *Human Summary* as an oracle query baseline, where expert-written policy summaries are used directly as retrieval queries; this condition is not reproducible at inference time since it requires human annotation, but it establishes an upper bound on the quality achievable when the query perfectly captures policy intent. The practical goal for automated methods is therefore to close the gap to this oracle ceiling.

SEEK-Policy achieves the strongest performance among all fully automatic methods, with Hit@1 = 0.370, NDCG@10 = 0.535, and MRR@5 = 0.474, recovering approximately 54% of the Hit@1 gap between the keyword baseline and the oracle ceiling (0.052 → 0.688).

The keyword-based *ClimatePolicyRadar* baseline performs poorly across all metrics (Hit@1 = 0.052, NDCG@10 = 0.124), confirming that lexical matching alone is insufficient for linking dynamic environmental signals to semantically relevant policy content.

Among the Siamese configurations, contrastive loss consistently and substantially outperforms triplet loss across backbones, with Hit@1 gains of 0.251 (ClimateBERT) and 0.221 (all-distilroberta) at $w = 10$. The two backbones perform comparably under contrastive loss (Hit@1 ≈ 0.304 for both), suggesting that the loss function is the dominant factor in Siamese retrieval quality rather than the choice of text encoder. Despite these improvements, the best Siamese configuration still lags behind SEEK-Policy on Hit@1 (0.304 *vs*. 0.370), indicating that abstracting time-series patterns into natural language before retrieval provides a meaningful advantage over direct embedding alignment.

The Role-Agent system (Hit@1 = 0.301, NDCG@10 = 0.449) performs comparably to the best Siamese configurations on Hit@1 but falls short of SEEK-Policy across all metrics. While role-conditioned prompting introduces diverse analytical perspectives, its outputs are descriptive rather than optimised for discriminative retrieval, which limits ranking precision.

SEEK-Policy outperforms all automatic baselines by converting time-series trajectories into a structured natural language query *via* RAG-based summarisation before semantic search, allowing the retrieval stage to operate entirely in text space. The remaining gap to the human ceiling (Hit@1: 0.370 *vs*. 0.688) reflects the inherent difficulty of replacing expert policy interpretation with automated time-series summarisation, and points to summary quality as the primary direction for future improvement.

To provide a clearer reliability narrative, we interpret the validation results as four complementary forms of evidence. First, the automatic retrieval metrics demonstrate ranking effectiveness against strong baselines. Second, summary-quality evaluation assesses whether generated queries preserve semantic alignment with expert-written descriptions. Third, ablation and prompt-stability analyses examine whether performance remains stable across temporal windows, table-selection strategies, and random seeds. Fourth, error analysis identifies the remaining failure modes in dense semantic retrieval. Together, these analyses show that SEEK-Policy is reliable at the system level while also revealing limitations related to hallucination, ranking calibration, and fine-grained policy relevance.

Summary quality evaluation

A key concern in retrieval pipelines that rely on generated queries is whether poor query quality propagates errors to the retrieval stage, a "garbage in, garbage out" failure mode. To address this, we evaluate SEEK-Policy-generated summaries against expert-written human summaries using both automatic metrics and an LLM-as-a-Judge protocol.

Automatic metrics (ROUGE and BERTScore)

Table 3 reports ROUGE (*Lin, 2004*) and BERTScore (*Zhang et al., 2020*) for generated summaries at $w = 10$. ROUGE scores are expectedly low (ROUGE-1 = 0.243, ROUGE-2 = 0.025, ROUGE-L = 0.127), as the generated summaries are derived from numerical time-series patterns rather than paraphrasing human-written text, making lexical overlap an inherently limited fidelity measure in this cross-modal setting. In contrast, BERTScore, which operates in semantic embedding space and captures meaning beyond surface form,

Table 3 Automatic summary quality metrics for SEEK-Policy generated queries against human summaries at $w = 10$.

Metric	Score
ROUGE-1	0.243
ROUGE-2	0.025
ROUGE-L	0.127
BERTScore	0.833

yields a strong F1 of 0.833, indicating that the generated summaries are semantically well-aligned with expert descriptions of the same policies. Full results across all window sizes are provided in 'Summary Quality Across Window Sizes', confirming that summary quality remains stable across $w \in \{1, 2, 5, 10\}$.

LLM-as-a-judge evaluation

To complement automatic metrics with a more nuanced assessment of summary quality, we conduct an LLM-as-a-Judge evaluation using GPT-4o as the judge model (*Gu et al., 2025*) using deepeval python library (http://deepeval.com/). We evaluate two dimensions: (1) Temporal-Semantic Alignment, which assesses whether the summary accurately reflects the quantitative trends (upward/downward/stable) present in the raw time-series data, and (2) Hallucination, which measures whether the summary introduces factual claims absent from the structured document metadata (title, geography, sector).

Evaluation is conducted on a stratified random sample of 100 queries ($n = 50$ retrieval successes, $n = 50$ retrieval failures), drawn with a round-robin geographic sampling strategy across five continents to ensure representative coverage of the POLiMATCH *corpus*. Both the SEEK-Policy-generated summaries and the human summaries are evaluated for the same documents, enabling direct comparison under identical judge conditions.

Table 4 reveals two complementary findings. First, SEEK-Policy achieves a higher temporal-semantic alignment score (0.459 *vs.* 0.342), reflecting that its summaries are generated directly from time-series data and therefore capture trend directions more faithfully than expert-written summaries, which describe policy intent rather than empirical observations. Second, SEEK-Policy exhibits a higher hallucination score (0.588 *vs.* 0.230), which is an expected characteristic of LLM-generated text and likely contributes to the remaining performance gap between SEEK-Policy and the human oracle baseline (Hit@1: 0.370 *vs.* 0.688). This suggests that reducing hallucination in the query generation stage is a promising direction for closing this gap in future work.

Qualitative example. Table 5 illustrates a representative case from the evaluation. For a Chilean energy policy document, SEEK-Policy generates a detailed summary describing the country's transition from fossil fuels towards renewable energy, which the judge scores with alignment = 0.767 while noting a minor hallucination (score = 0.20) for omitting the specific law number. The human summary, by contrast, is highly precise about the legal instrument but does not capture the underlying empirical trends. This example illustrates

Table 4 LLM-as-a-Judge evaluation comparing SEEK-Policy generated summaries and human summaries. Alignment: higher is better (GEval, 0-1). Hallucination: lower is better (Hallucination-Metric, 0-1). Bold values indicate the better-performing result for each metric.

Summary type	Alignment ↑	Hallucination ↓
Human summary	0.342	**0.230**
SEEK-Policy	**0.459**	0.588

Table 5 Qualitative LLM-as-a-Judge example: chile energy policy (Law 21,118). Judge model: GPT-4o.

Type	Judge reasoning (alignment)	Align.	Halluc.
SEEK-Policy	Accurately describes Chile's energy transition towards renewables and legislative measures; minor incompleteness on specific 2035/2050 targets.	0.767	0.200
Human	Precisely identifies the legal instrument and capacity increase to 300 MW, but does not reflect empirical time-series trends.	0.281	0.000

the complementary strengths of the two summary types and confirms that SEEK-Policy's hallucinations are largely peripheral rather than substantive.

Ablation study

Effect of historical window size

Table 6 reports SEEK-Policy performance across historical window sizes $w \in \{1, 2, 5, 10\}$. Results are stable across all four horizons, with absolute differences of at most 0.015 in Hit@1 and 0.009 in NDCG@10, indicating that the system is robust to window length choice. The longest window ($w = 10$) achieves the best or joint-best score on all reported metrics, suggesting that broader temporal context provides a marginal but consistent benefit for retrieval. This pattern is consistent across all methods (see 'Full Window Results Across All Methods'), further supporting the conclusion that window length does not distort retrieval quality. Notably, this finding also provides empirical support for the zero-padding strategy applied to countries with fewer than w years of observations. If padding were harmful, performance would degrade monotonically with w since longer windows require more padding by construction. The absence of such degradation indicates that zero-padded entries do not adversely affect retrieval quality.

LLM-based vs. rule-based table selection

To assess whether the LLM-driven table selection stage contributes meaningfully to retrieval quality, we compare it against a rule-based baseline that replaces the LLM with an embedding similarity approach. Specifically, the baseline encodes the dataset description and the metadata of each candidate table using a fixed-sentence encoder, computes cosine similarity between the two, and selects at most $t = 5$ tables per query. This process requires no language model reasoning and relies purely on surface-level semantic overlap between the query description and table metadata. Both configurations are evaluated using the *SEEK-Policy* retrieval pipeline with a window size $w = 10$ across five cross-validation folds.

Table 7 reports the results. The LLM-based selection outperforms the rule-based baseline across all metrics. Hit@1 improves from 0.355 to 0.370 (+1.5). Ranking quality

Table 6 Ablation on historical window size for SEEK-Policy (rag_summary_semantic). Best result per metric in bold.

Window	Hit@1	NDCG@5	NDCG@10	MRR@5
$w = 1$	0.368	0.515	0.534	**0.475**
$w = 2$	0.361	0.506	0.530	0.466
$w = 5$	0.355	0.507	0.528	0.465
$w = 10$	**0.370**	**0.515**	**0.535**	0.474

Table 7 Comparison of LLM-based and rule-based table selection strategies using SEEK-Policy ($w = 10$, 5-fold cross-validation). Bold values indicate the best-performing result for each metric. Italicized values indicate the absolute performance difference between the LLM-based and rule-based strategies (LLM-Rule).

Selection strategy	Hit@1	NDCG@5	NDCG@10	MRR@5
Rule-based (embedding similarity)	0.355	0.503	0.521	0.462
LLM-based (proposed)	**0.370**	**0.515**	**0.535**	**0.474**
Δ *(LLM − Rule)*	*+0.015*	*+0.012*	*+0.014*	*+0.012*

follows the same trend, with NDCG@10 increasing from 0.521 to 0.535 and MRR@5 increasing from 0.462 to 0.474. Although the absolute gains are modest, they are consistent across all evaluated metrics and retrieval depths, suggesting that the LLM contributes genuine reasoning beyond what can be recovered from metadata similarity alone.

Prompt stability

Since the SEEK policy involves multiple LLM prompting stages, non-deterministic outputs across runs are an inherent concern. To assess this, we ran SEEK-Policy on 20 stratified samples across three random seeds (42, 123, 777) and measured variance in retrieval performance. Results confirm that some degree of variability exists, particularly at Hit@1 (CV = 19.29%), reflecting the sensitivity of top-1 correctness to small changes in generated summaries. However, ranking-based metrics remained relatively stable, with NDCG@10 CV of 4.92% and MRR@5 CV of 6.86%, suggesting that while individual top-1 predictions may shift across runs, the overall ranking quality is consistent. Users of SEEK-Policy should therefore treat top-1 results with appropriate caution and consider top-5 or top-10 recommendations for more reliable decision support. Full results are reported in 'Prompt Stability Results'.

Error analysis

To complement the performance evaluation, we analyze the error patterns of the retrieval models by examining the cosine similarity distributions between queries and retrieved documents. This analysis focuses on three representative systems: SEEK-Policy, the Role-Based Multi-Agent model, and the Siamese Network Model. Figure 6 illustrates the distributions of similarity scores for correct predictions (red) and incorrect predictions (green).

For SEEK-Policy, correct predictions concentrate in the moderate-to-high similarity range, peaking around 0.65–0.68, while incorrect predictions are skewed toward higher

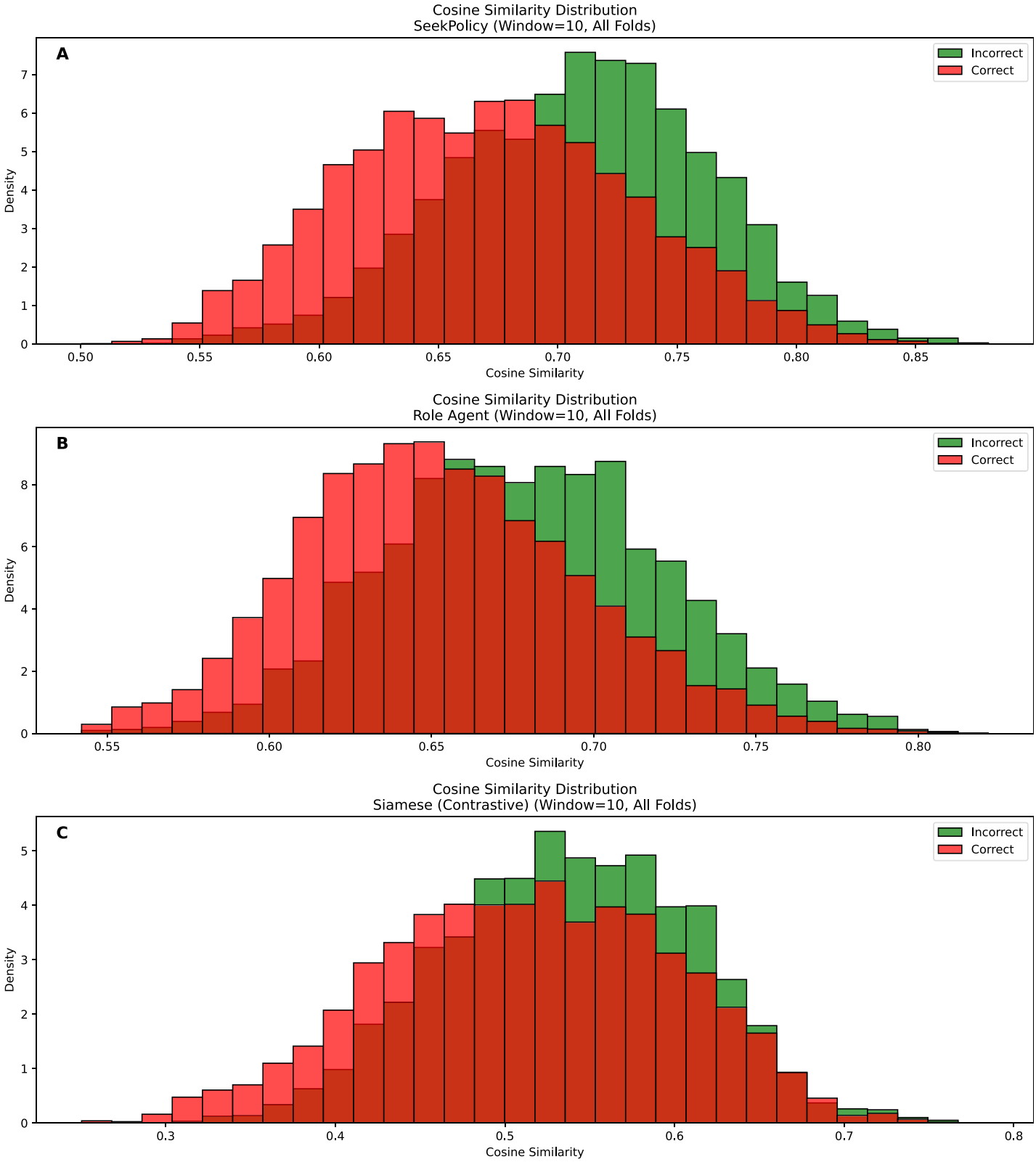


Figure 6 (A–C) Cosine similarity distributions for correct and incorrect predictions across models. Green bars indicate correct predictions and red bars indicate incorrect predictions.

Full-size DOI: 10.7717/peerj-cs.4054/fig-6

values, clustering between 0.70 and 0.85. This rightward shift of incorrect predictions reveals a systematic tendency for irrelevant documents to receive inflated similarity scores, creating a band of overlap in the 0.68–0.75 range. While SEEK-Policy achieves reasonable retrieval performance overall, this pattern suggests that the embedding space still conflates certain irrelevant passages with relevant ones at higher similarity levels. That additional ranking calibration could reduce this confusion.

The Role-Based Multi-Agent model displays a similar but more pronounced pattern. Correct predictions peak around 0.63–0.65, yet incorrect predictions are concentrated at notably higher similarities, between 0.70 and 0.80, with a long tail extending toward 0.85. This inversion, where incorrect retrievals receive higher scores than correct ones, indicates that role-conditioned representations, while semantically rich, do not reliably discriminate relevant from irrelevant documents. The model's embeddings appear to reward surface-level or topical similarity rather than precise policy relevance, leading to systematic over-retrieval of loosely related passages.

The Siamese Network Model exhibits the weakest separation across all three systems. Both correct and incorrect predictions overlap broadly in the 0.35–0.65 range, with peaks around 0.50–0.53 for correct predictions and a slight rightward lean for incorrect predictions at higher similarities (0.55–0.65). The near-complete overlap between the two distributions confirms that the model cannot reliably distinguish relevant from irrelevant documents, and its latent space provides little discriminative signal for ranking. This poor separation reflects the fundamental difficulty of aligning raw time-series embeddings with unstructured policy text using contrastive training alone.

Insights. Across all three models, the error distributions reveal a shared challenge: incorrect predictions tend to cluster at mid-to-high similarity scores, creating overlap zones that undermine retrieval precision. Notably, both the SEEK-Policy and the Role-Based Multi-Agent model exhibit a counterintuitive pattern where incorrect retrievals receive higher similarity scores than correct ones in certain ranges, suggesting that their embedding spaces capture broad semantic proximity but struggle to encode fine-grained policy relevance. The Siamese model compounds this issue with an almost undifferentiated similarity distribution, reflecting the limitations of contrastive objectives when applied across heterogeneous modalities. Addressing these patterns in future work will require improved calibration strategies, modality-aware embedding alignment, and potentially hybrid ranking approaches that combine dense retrieval with complementary signals.

Case study

To further illustrate the effectiveness of SEEK-Policy, we conducted a case study comparing its top-1 retrieval with the keyword-based search in Climate Policy Radar. Figure 7 presents an example where the query "Energy Demand Israel" fails under keyword search. Because the target policy text does not contain the exact word "Israel," the correct document is missed, while irrelevant documents containing repeated terms such as "energy" are overemphasized. This reflects the rigid dependence of keyword search on surface-level term matching.

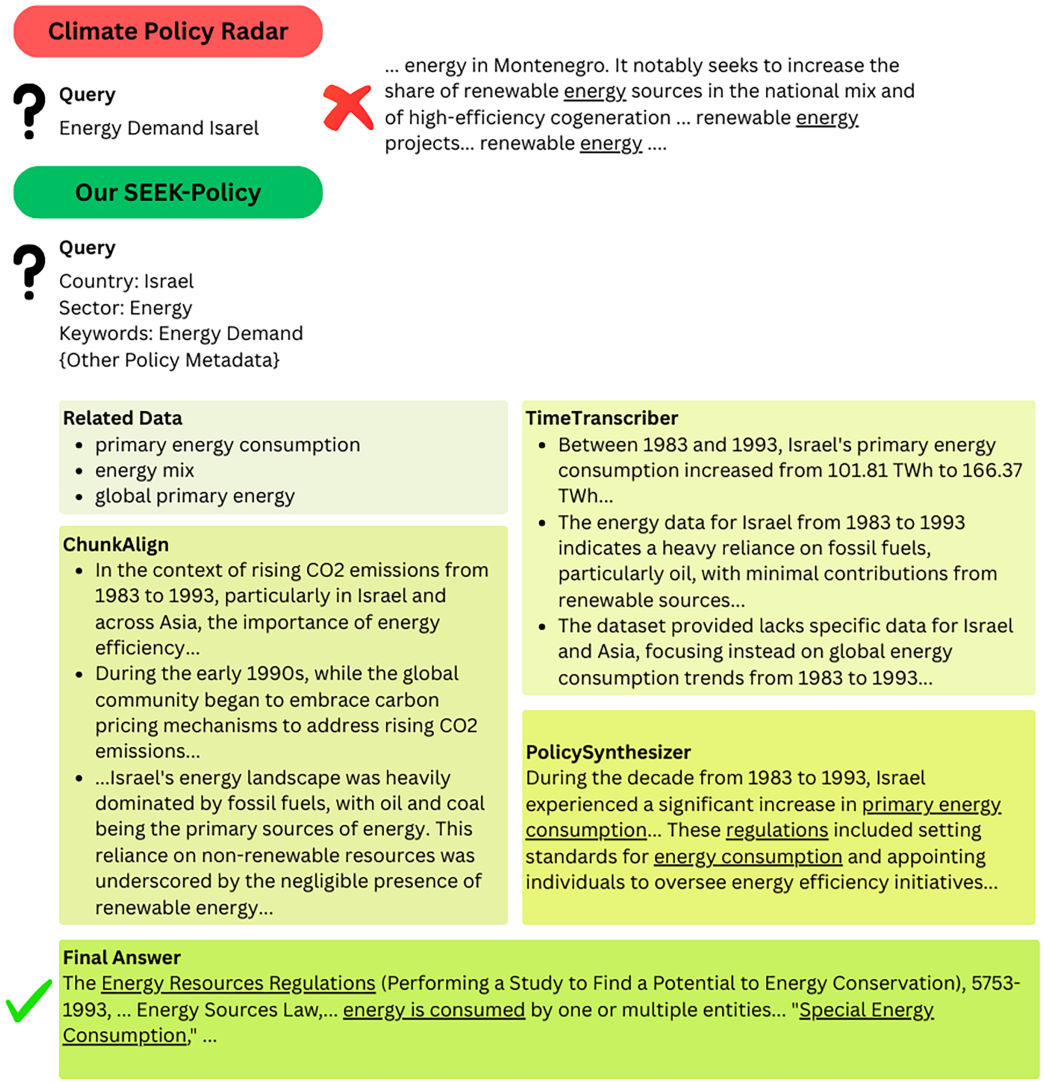


Figure 7 Case study comparing SEEK-Policy and keyword-based search. While keyword search fails to retrieve the correct Israeli energy policy when exact terms are missing, SEEK-Policy identifies the relevant regulation by combining metadata-driven table selection, semantic chunk retrieval, and policy-oriented synthesis. Full-size DOI: 10.7717/peerj-cs.4054/fig-7

SEEK-Policy overcomes this limitation by combining metadata and semantic retrieval. First, Related Data tables are identified using policy metadata (country = Israel, sector = energy, keywords = energy demand). Second, the TimeTranscriber module transforms these time-series indicators into a concise natural language summary, capturing trends in Israel's primary energy consumption between 1983 and 1993. Third, the *ChunkAlign* retriever locates semantically aligned passages within policy documents, even when wording differs; in this case, it aligns "energy demand" with mentions of "energy consumption" and "primary energy use." Finally, the PolicySynthesizer composes the retrieved passages into a coherent output that contextualizes Israel's fossil fuel reliance, efficiency measures, and relevant regulatory actions. This pipeline enables the system to

correctly identify the regulation titled Energy Resources Regulations, which keyword search was unable to retrieve.

Insights. This case study highlights two critical strengths of SEEK-Policy. First, the use of time-series-driven summaries ensures that retrieval is grounded in the actual energy trends of the country in question. Second, semantic chunk-level retrieval allows the system to generalize across related terms and avoid the pitfalls of exact keyword dependence. Together, these features enable SEEK-Policy to retrieve contextually relevant climate policies even when user queries do not perfectly match document wording.

DISCUSSION

Climate policy evolution and system-level implications

To situate the contributions of SEEK-Policy within the broader context of climate governance, it is necessary to examine how climate policies have developed and evolved over time. Drawing on large-scale evidence from the *ScanClimatePolicy* dataset, which analyzes more than 6,000 policies across 198 countries from 1924 to 2024, several patterns can be identified that inform both the architecture of contemporary climate policy and the design principles underlying our framework. Sectoral heterogeneity remains a defining characteristic of global climate governance. High-emission domains such as energy production, transportation, and buildings rely predominantly on structured targets and economic instruments, including renewable energy mandates, carbon pricing schemes, and fiscal incentives that facilitate large-scale decarbonization. By contrast, sectors marked by ecological complexity and more diffuse governance, such as agriculture and forestry, tend to emphasize regulatory measures and direct support programs. This stratification highlights the importance of context-sensitive alignment and directly motivates SEEK-Policy's sector-specific filters, which adapt recommendations to operational and institutional conditions.

The temporal evolution of policy portfolios further demonstrates the diversification of climate governance strategies. As illustrated in Fig. 8, diversification accelerated in the early 2000s, coinciding with international milestones such as the Kyoto Protocol and the Paris Agreement. Since that period, research, development, and innovation (RDI) instruments have been increasingly integrated alongside regulatory frameworks and strategic planning measures. This development reflects a broader transition toward adaptive and innovation-driven governance models that are capable of responding to accelerating climate risks and emerging technological opportunities. These dynamics have direct implications for SEEK-Policy. By semantically linking multivariate, real-time time-series data with historically grounded precedents, the framework is able to provide recommendations that are both context-aware and structurally coherent. For example, when environmental indicators reveal rapid increases in emissions or energy demand, SEEK-Policy can identify precedent cases where governments combined regulatory instruments with RDI measures, a pattern that has become particularly prominent in post-2000 energy and industrial strategies.



Figure 8 Proportion of policy instruments over time across different sectors. The heatmap illustrates the relative proportion of various policy instruments such as economic instruments, regulatory instruments, information and education, research, development, and innovation, and support and strategic measures—employed in different sectors over time. The sectors analyzed include (A) energy production and supply, (B) industrial processes, (C) transportation, (D) buildings and infrastructure, and (E) agriculture and forestry, spanning the years 1927 to 2023.

Full-size DOI: 10.7717/peerj-cs.4054/fig-8

Integrating these longitudinal and cross-sectoral insights strengthens SEEK-Policy's capacity to generate empirically grounded and context-aware recommendations. The multi-level retrieval filters incorporated into the framework ensure that outputs are not only semantically aligned with environmental data but also consistent with the sectoral, geographic, and institutional features of the policy domain. This design enhances interpretability, reduces the cognitive burden on policymakers, and facilitates more

efficient policy synthesis. Taken together, these insights highlight a central contribution of our work: SEEK-Policy demonstrates how dynamic environmental indicators can be systematically bridged with historically validated policy strategies, offering a scalable, evidence-based, and adaptive framework for climate policy recommendation that extends beyond the capabilities of existing systems.

Practical limitations and deployment considerations

Beyond these methodological limitations, several practical considerations should be addressed before deploying SEEK-Policy in real-world policy settings. First, multilingual applicability remains a constraint because the current evaluation relies primarily on English-language policy summaries, whereas climate policy documents are produced in diverse languages, legal styles, and administrative formats. Although the framework can be extended with multilingual encoders or cross-lingual embedding models, retrieval quality may vary across low-resource languages and should be validated by domain experts before operational use.

Second, policy-domain bias may arise from uneven coverage across countries, sectors, time periods, and policy instruments. Regions with more complete English metadata or stronger documentation infrastructures may be over-represented, while under-documented domains such as agriculture, forestry, local adaptation, and informal governance may receive weaker retrieval coverage. As a result, recommendations may reflect data-rich jurisdictions or dominant policy domains rather than the full diversity of feasible policy options in resource-constrained contexts.

Third, operational use would require continuous data updating, version control for evolving policy documents, periodic embedding refresh, data-quality monitoring, computational resources, and clear governance procedures. Because SEEK-Policy includes LLM-based components, its outputs may also contain incomplete, overly general, or hallucinated statements, consistent with the higher hallucination score observed in the LLM-as-a-Judge evaluation. Therefore, SEEK-Policy should be used as a decision-support tool for evidence discovery and comparative policy analysis, rather than as an automated policy-making system. Final interpretation and adoption of recommendations should remain under the supervision of policy experts and institutional stakeholders.

CONCLUSION AND FUTURE WORK

This article introduced SEEK-Policy, a framework for climate policy recommendation that integrates multivariate time-series data with policy document retrieval through a cross-modal, retrieval-augmented generation architecture. To support this task, we developed the POLiMATCH dataset, which links structured environmental and socio-economic indicators with a curated collection of climate policy documents across multiple countries and sectors, providing a foundation for evaluating context-aware retrieval systems.

Experiments demonstrated that SEEK-Policy consistently outperforms the baselines evaluated, including a Siamese Network model trained with contrastive loss and a

Role-Based Multi-Agent approach. The framework achieved higher accuracy across all retrieval metrics, and error analysis confirmed stronger separation between correct and incorrect matches compared with alternative models. These results support the utility of converting time-series signals into natural language before semantic retrieval, rather than aligning raw numerical embeddings directly with policy text.

Nevertheless, a notable performance gap persists between SEEK-Policy and the human-oracle baseline (Hit@1: 0.370 *vs.* 0.688). Our LLM-as-a-Judge evaluation suggests that while SEEK-Policy summaries capture temporal trends more faithfully than human-written summaries (alignment: 0.459 *vs.* 0.342), they also exhibit higher hallucination (0.588 *vs.* 0.230), which likely accounts for much of this gap. Reducing hallucination in the query generation stage, therefore, represents the most critical direction for improvement.

Several limitations also remain. The LLM components rely on general-purpose models without domain-specific fine-tuning, as adapting a language model to the climate policy domain would require substantial labeled data and computational resources beyond the scope of this work. Fine-tuning a domain-adapted encoder remains a clear direction for future work. Additional extensions include integrating time-series-specific architectures such as TimeLLM (*Jin et al., 2024a*), incorporating complementary modalities such as satellite imagery and macroeconomic indicators, and expanding POLiMATCH to cover more languages and regions to improve fairness and coverage.

In summary, SEEK-Policy offers a practical approach to linking environmental time-series data with climate policy documents, and the POLiMATCH dataset provides a reusable benchmark for this task. As climate governance becomes more complex, tools that connect real-time environmental signals with relevant policy evidence offer a promising step toward more timely, context-aware decision support.

ETHICS DECLARATIONS

This study uses only publicly available, open-access datasets and does not involve human subjects or personal data. However, several ethical considerations warrant disclosure. First, the reliance on English-language policy summaries may introduce systematic bias against policies from the Global South, where documentation is often available only in local languages, potentially under-representing these regions in retrieval outcomes. Second, the underlying LLMs may reflect biases in their training corpora, such as over-representation of policy instruments from high-income countries, which could skew recommendations toward approaches that are less feasible in resource-constrained contexts. We have not conducted formal toxicity tests or fairness audits across demographic or regional subgroups, and we acknowledge this as a direction for future work. Finally, SEEK-Policy is intended strictly as a decision-support tool to assist policymakers in identifying relevant evidence, and should not be interpreted as an autonomous decision-maker. All outputs should be reviewed critically by domain experts before informing policy action.

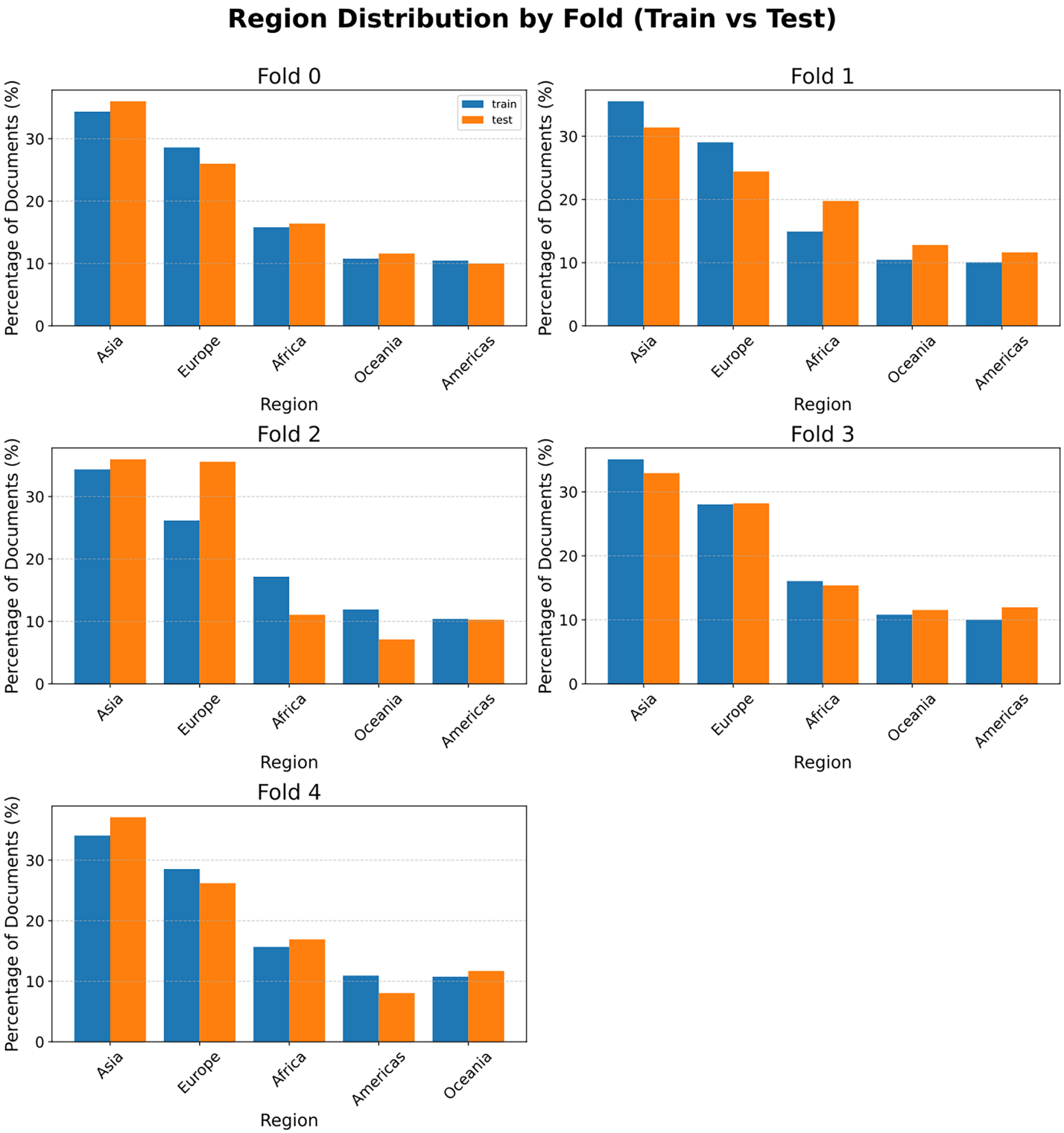


Figure 9 **Regional distribution of policy documents across the five cross-validation folds, ensuring balanced representation in each test split.**
Full-size DOI: 10.7717/peerj-cs.4054/fig-9

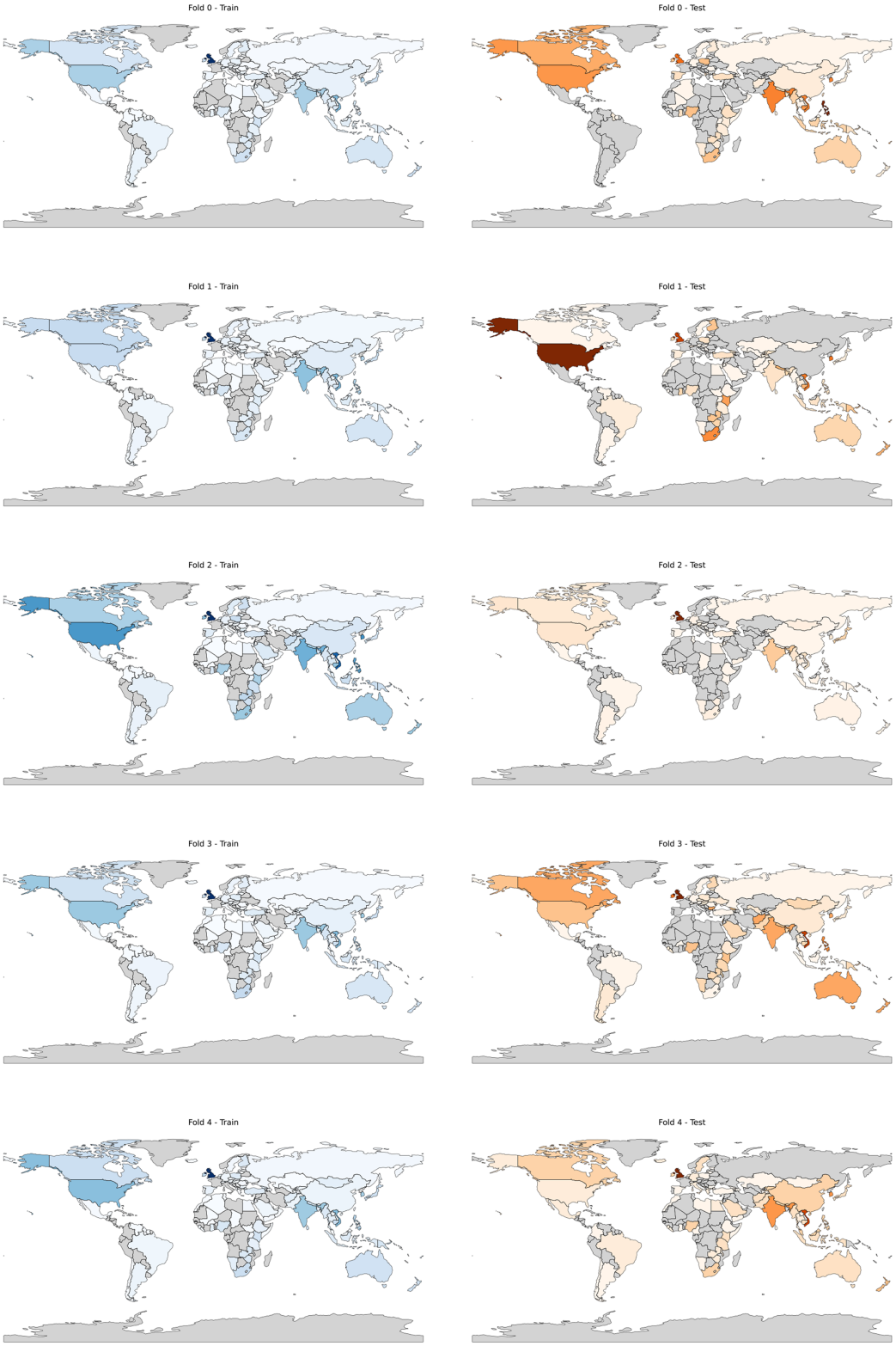


Figure 10 **Country distribution of policy documents across the five cross-validation folds, ensuring balanced representation in each test split.** Full-size DOI: 10.7717/peerj-cs.4054/fig-10

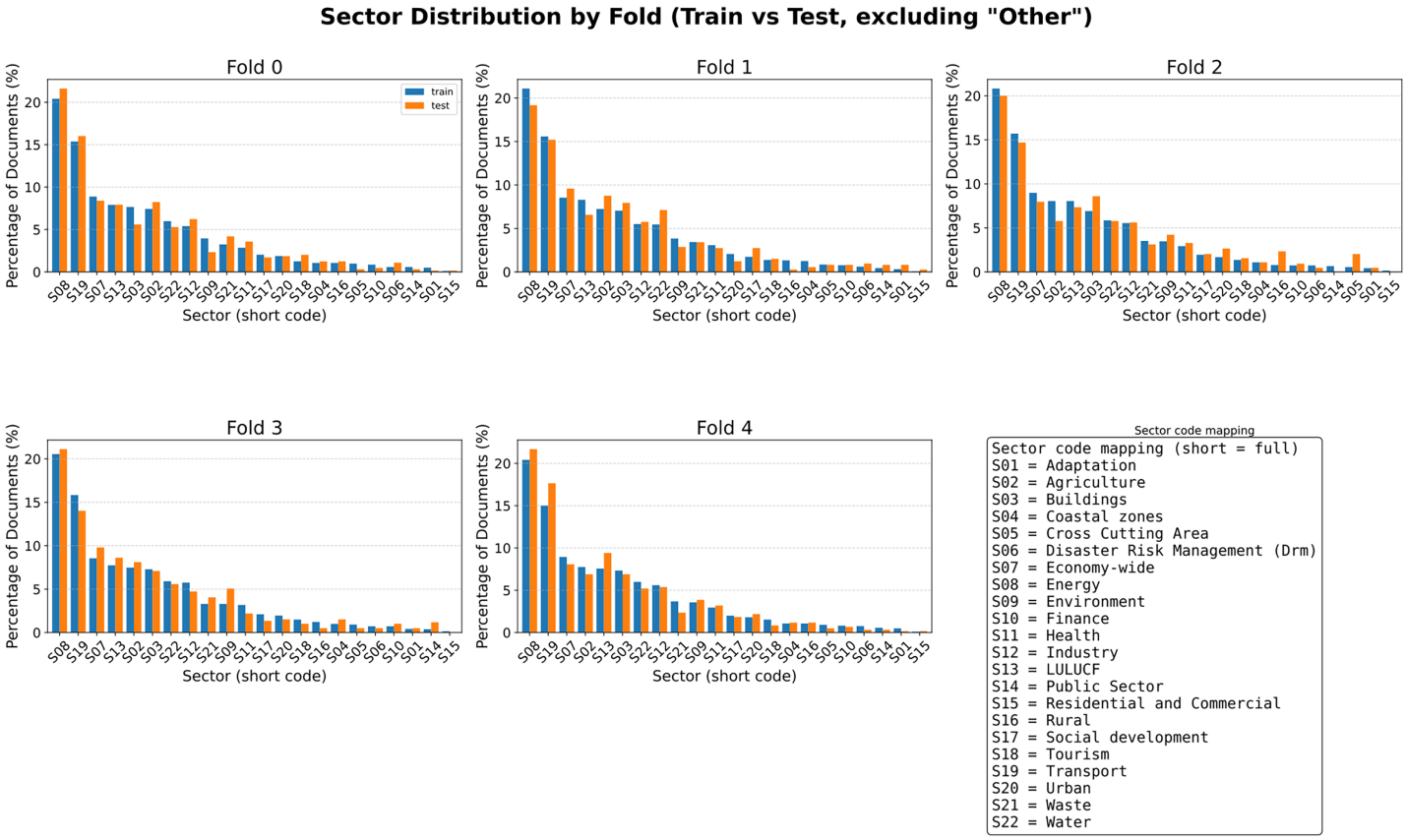


Figure 11 **Sector distribution of policy documents across the five cross-validation folds, ensuring balanced representation in each test split.**
Full-size DOI: 10.7717/peerj-cs.4054/fig-11

DATASET DISTRIBUTION ACROSS CROSS-VALIDATION FOLDS

To ensure the robustness of the Group K-Fold cross-validation, we analyzed the consistency of the data distribution across all five folds. Figures 9, 10, and 11 illustrate the regional, country-level, and sectoral breakdown for each fold. The regional distributions show a consistent representation of all five geographic areas—Asia, Europe, Africa, Oceania, and the Americas—across both the train and test splits of every fold. By maintaining a balanced distribution of geographic regions and policy sectors across all folds, we mitigate the risk of performance bias arising from uneven data coverage. This consistency ensures that the improvements observed in SEEK-Policy are statistically representative of the entire POLiMATCH *corpus* rather than a result of any specific, high-frequency data subset.

PROMPTS FOR ROLE-BASED AGENTS APPROACH

In this section, we present the prompts used in the Role-Based Agents approach for generating policy-relevant summaries. Each prompt corresponds to a specific role (*e.g.*, climate policy advisor, socio-economic policy advisor) and guides the LLM in generating sector-specific insights from the time-series data.

Prompt for climate policy advisor

You are a climate policy and environmental data advisor tasked with conducting a sector-specific analysis of the provided climate-related dataset. Your goal is to extract meaningful insights, focusing on trends, anomalies, and patterns within the specified sector(s), which will inform policymakers on effective strategies to mitigate climate change impacts. Use the data to identify critical factors such as greenhouse gas emissions, deforestation, ocean heat content, and other relevant indicators based on the sector(s). Highlight correlations, year-on-year changes, and significant shifts that could impact future policies in the user-defined sector(s).
Key Instructions:
1. **Analyze Sector-Specific Trends:**
- Examine trends relevant to the **specified sector(s)**, such as changes in emissions, energy consumption, deforestation, or wildfire emissions, as applicable.
- Investigate changes in greenhouse gas emissions (*e.g.*, CO_2, methane, nitrous oxide) across relevant activities within the sector(s) (*e.g.*, energy, manufacturing, or other user-defined sectors).
- Track how emissions per unit of GDP, *per capita*, or other relevant economic indicators have evolved over time, identifying sector-specific patterns or shifts.
2. **Correlations & Relationships within the Sector(s):**
- Explore correlations between key metrics within the **sector(s)**, such as the relationship between deforestation rates and emissions, or energy consumption and carbon sequestration, depending on the sector(s).
- Analyze any sector-specific connections between environmental variables like ocean heat content and wildfire emissions, or between intact forest landscapes and carbon sequestration.
- Investigate other relationships that may influence the environmental footprint of the sector(s), such as resource extraction or land-use changes.
3. **Anomalies and Shifts:**
- Identify any significant anomalies in the data within the sector(s), such as unusual spikes in emissions, deforestation rates, or energy consumption in specific years or regions.
- Analyze sudden shifts in key metrics like emissions intensity, forest cover, or energy demand within the sector(s), providing possible explanations and implications for policy adjustments.
4. **Data-driven Policy Recommendations for the Sector(s):**
- Based on the analysis, offer **targeted policy recommendations** aimed at reducing emissions, energy consumption, or deforestation specific to the user-defined sector(s).
- Prioritize actions in regions or sub-sectors where environmental risks (*e.g.*, deforestation, wildfire) are high, proposing interventions that mitigate these effects within the sector(s).

- Highlight sustainable practices, clean technologies, or policy instruments that could help reduce emissions per unit of GDP or energy in the sector(s), supporting economic growth without sacrificing environmental goals.
Additional Considerations:
1. Focus on **temporal dynamics** specific to the sector(s)—how have emissions, energy usage, and environmental damage evolved year-over-year within the sector (s)?
2. Provide a **region-specific breakdown** within the sector(s) where applicable, focusing on areas of the highest environmental impact.
3. Offer actionable insights on how **policymakers** can reduce emissions, deforestation, or energy demand in the sector(s) while balancing economic considerations.
**Targeted Sector(s): ** {sectors}
Data:
{climate_data}

Prompt for socio-economic advisor

You are a socio-economic policy expert tasked with analyzing socio-economic data from the past years, with a focus on one or more specified sectors or regions. Your goal is to provide a comprehensive analysis of the data, focusing on trends, anomalies, and patterns that are crucial for policymakers to design and implement effective socio-economic policies. Use the provided data to identify key factors affecting economic growth, inequality, labor markets, and poverty within the specified sectors or regions. Highlight any correlations, year-on-year changes, or regional disparities that could impact future policy decisions.
Key Instructions:
1. **Analyze Sector- or Region-Specific Trends:**
- Identify major trends in GDP growth, employment rates, income inequality, and poverty reduction over the years, specific to the user-defined sector(s) (*e.g.*, healthcare, education) or region(s) (*e.g.*, urban *vs.* rural areas, developing *vs.* developed economies).
- Examine how these trends influence various socio-economic outcomes in the selected sectors or regions, and highlight any disparities.
2. **Correlations & Relationships within the Sector(s) or Region(s):**
- Look for correlations between socio-economic indicators in the specified sector(s) or region(s), such as the relationship between income inequality and access to education, or between urbanization and unemployment rates.

- Explore how socio-economic events (*e.g.*, recessions, economic booms) have impacted poverty levels, income distribution, and labor markets in the user-specified sectors or regions.
3. **Anomalies and Shifts:**
- Identify any significant anomalies (*e.g.*, sudden drops or spikes) in the socio-economic data specific to the sector(s) or region(s), and provide potential explanations for these changes.
- Analyze sudden shifts in economic trends, such as sharp increases in unemployment or income inequality, and suggest how these shifts could guide future policy decisions for the sector(s) or region(s).
4. **Data-Driven Policy Recommendations:**
- Based on your analysis, provide **targeted policy recommendations** for improving socio-economic outcomes within the defined sector(s) or region(s).
- Prioritize areas that show significant disparities or rapid economic changes, and propose policies (*e.g.*, tax reforms, social safety nets, education programs) that align with the specific socio-economic needs of the sector(s) or region(s).
- Highlight inclusive growth strategies or instruments that can reduce inequality and improve living standards in marginalized populations or underdeveloped areas.
Additional Considerations:
1. Focus on **temporal dynamics**—how have socio-economic indicators evolved year-over-year in the specified sector(s) or region(s)?
2. Provide a **region- or sector-specific breakdown** where applicable, focusing on disparities and critical factors affecting policy outcomes.
3. Offer insights on how **policymakers** can promote inclusive growth, reduce inequality, and improve living standards in underdeveloped areas or among marginalized populations, especially within the user-defined scope.
**Targeted Sector(s): ** {sectors}
Data:
{socio_economic_data}

Prompt for sustainable development and public policy advisor

You are a **sustainable development and public policy advisor** specializing in the analysis of **socio-economic, environmental, and infrastructure-related data**. Your objective is to evaluate the provided dataset, which covers the **sectors**. Your goal is to generate comprehensive insights and **policy recommendations** that balance **economic growth, environmental sustainability, and public well-being**. These insights will assist policymakers in implementing targeted interventions specific to the **{sectors}**.
Key Instructions:

1. **Sector-Specific Trend Analysis:**
- Analyze trends in **{sectors}** such as **sector-specific metrics, *e.g.*, energy production, emissions, agriculture, transportation, *etc.*** and assess how changes impact the economy and environmental sustainability within these sectors.
- Focus on detailed metrics related to **sector-specific issues, *e.g.*, energy mix, emissions *per capita*, agriculture yields, transportation emissions, public health metrics, *etc.***, linking these patterns to their broader socio-economic effects.
2. **Correlations Between Key Indicators:**
- Investigate correlations between **sector-specific indicators** and environmental factors, such as the impact of **energy production on sustainability targets, or transportation policies on emissions and public health outcomes**.
- Explore the interconnection between **sector-specific policies** (*e.g.*, **regulatory changes, economic incentives, public infrastructure projects**) and **socio-economic indicators** (*e.g.*, GDP, public health metrics, environmental sustainability).
3. **Anomalies and Significant Shifts:**
- Identify significant anomalies within **sectors**, such as abrupt changes in production, consumption patterns, or emissions. Explore the socio-economic and environmental drivers behind these shifts, with a particular focus on **sector-specific anomalies, *e.g.*, energy, transportation, public health, agriculture**.
- Highlight any significant spikes or reductions in **sector-specific metrics** (*e.g.*, transportation emissions, energy consumption, public health indicators), providing context for these shifts and proposing actions to mitigate adverse impacts.
4. **Cross-Sector Policy Recommendations:**
- Provide **integrated policy recommendations** that address specific challenges in the **sectors**, balancing **economic growth, public well-being, and environmental sustainability**. Focus on **sector-specific improvements, *e.g.*, energy efficiency, emission reduction, sustainable agricultural practices**.
- Propose measures to enhance **public health outcomes, infrastructure development**, or **environmental sustainability** specific to **{sectors}**, ensuring a targeted approach to addressing sector-specific challenges.
Additional Considerations:
1. Analyze how trends in **{sectors}** evolve over time and vary across regions, providing region-specific insights where applicable.
2. Evaluate the effectiveness of current policies in **{sectors}** and highlight opportunities for improvement.
3. Offer region-specific recommendations that focus on **urgent actions** or **opportunities for sustainable development** within **{sectors}**.
Targeted Sector(s): {sectors}
Data:
{other_data}

Prompt for interdisciplinary climate policy expert

You are an interdisciplinary climate policy expert, specializing in analyzing data from climate experts, sector-specific advisors, and socio-economic advisors to propose climate-related policies for **{country}, {region}** within the applied **{sectors}**. Your task is to synthesize data and provide a climate-related policy recommendation, addressing both environmental challenges and socio-economic concerns.

—

Scenario Start:

You are provided with the following data from three expert advisors:

1. **Climate Expert**: Climate data that outlines trends, environmental impacts, and key sustainability challenges.
2. **Sector-Specific Advisor**: Data focused on specific sectors such as energy, transportation, or agriculture, highlighting current challenges and opportunities for improvement.
3. **Socio-Economic Advisor**: Data that analyzes socio-economic factors, helping evaluate the feasibility and impact of proposed policy interventions on public well-being and economic growth.

Using this information, propose a comprehensive climate-related policy for **{country}** in the **{sectors}**. Your response should include:

1. **Causes/Issues:** Identify the primary climate and socio-economic problems the policy needs to address based on the provided data.
2. **Targets/Goals:** Set clear objectives for the policy that promote environmental sustainability, economic growth, and public well-being.
3. **Policy Instruments:** Recommend specific tools or methods (*e.g.*, tax incentives, regulatory changes, infrastructure development) to achieve these goals within the applied sectors.
4. **Keywords:** Highlight key terms and concepts relevant to the proposed policy.

—

Data Provided:

- **Climate Data:** {climate_data}
- **Sector-Specific Data:** {other_data}
- **Socio-Economic Data:** {socio_economic_data}

—

End Scenario.

Prompt for policy summarizer

Task: You are an expert in summarizing policy analysis. Your role is to condense detailed input from a climate policy advisor into a clear, natural-flowing single-paragraph summary of the proposed policy.

Instructions: Summarize the policy with a natural and coherent flow. While the summary should include the identified causes or issues, main targets, recommended policy instruments, and important keywords, ensure the paragraph reads smoothly as a cohesive description rather than directly listing these elements. The goal is to provide a well-rounded, insightful summary without unnecessary repetition.
—
Begin!
Input from Policy Advisor:
{advisor_output}
Summary:

Table 8 Summary quality across window sizes. R-1 = Rouge-1, R-2 = Rouge-2, R-L = Rouge-L and B = BERTScore.

Window	R-1	R-2	R-L	B
$w = 1$	0.245	0.025	0.128	0.833
$w = 2$	0.243	0.025	0.127	0.833
$w = 5$	0.244	0.025	0.128	0.833
$w = 10$	0.243	0.025	0.127	0.833

Table 9 Full retrieval results across all methods and window sizes $w \in \{1, 2, 5, 10\}$. Metrics reported: Hit@1 (H@1), NDCG@10 (N@10), and MRR@5 (M@5). The bold method name denotes the proposed SEEK-Policy framework, while bold metric values indicate the results of the selected configuration, $w = 10$, used in subsequent experiments.

Method	Loss	$w = 1$			$w = 2$			$w = 5$			$w = 10$		
		H@1	N@10	M@5	H@1	N@10	M@5	H@1	N@10	M@5	H@1	N@10	M@5
Siamese (ClimateBERT)	Contrastive	0.291	0.518	0.423	0.292	0.525	0.424	0.290	0.512	0.415	0.304	0.533	0.438
Siamese (ClimateBERT)	Triplet	0.077	0.213	0.143	0.060	0.175	0.111	0.058	0.179	0.112	0.053	0.169	0.108
Siamese (all-distilroberta)	Contrastive	0.289	0.518	0.423	0.312	0.538	0.446	0.306	0.527	0.437	0.304	0.531	0.440
Siamese (all-distilroberta)	Triplet	0.113	0.267	0.189	0.081	0.236	0.160	0.085	0.224	0.152	0.083	0.220	0.147
Role-Agent (chunk semantic)	—	0.287	0.439	0.385	0.293	0.445	0.390	0.297	0.450	0.396	0.301	0.449	0.395
SEEK-Policy	—	0.368	0.534	0.475	0.361	0.530	0.466	0.355	0.528	0.465	**0.370**	**0.535**	**0.474**

SUMMARY QUALITY ACROSS WINDOW SIZES

Table 8 reports ROUGE and BERTScore for SEEK-Policy generated summaries across all historical window sizes. Results are highly consistent, with BERTScore F1 varying by less than 0.001 across all configurations, indicating that the quality of the generated query is not sensitive to the length of the historical window.

FULL WINDOW RESULTS ACROSS ALL METHODS

Table 9 reports retrieval performance for all methods across all historical window sizes $w \in \{1, 2, 5, 10\}$. The results confirm that the pattern observed for SEEK-Policy in Table 6

Table 10 Prompt stability across three random seeds (20 samples).

Metric	Seed 42	Seed 123	Seed 777	Mean	Std	CV (%)
Hit@1	0.380	0.260	0.360	0.333	0.064	19.29
NDCG@5	0.447	0.424	0.468	0.446	0.022	4.89
NDCG@10	0.488	0.443	0.474	0.468	0.023	4.92
MRR@5	0.429	0.384	0.437	0.417	0.029	6.86

is consistent across methods: performance differences across window sizes are small in all configurations, and no method exhibits a systematic degradation at $w = 10$ despite its higher reliance on zero-padding. This provides additional evidence that zero-padding does not introduce distortion into the retrieval process.

PROMPT STABILITY RESULTS

Table 10 reports retrieval metrics across three random seeds on 20 stratified samples, evaluating the stability of SEEK-Policy's LLM-generated summaries.

ADDITIONAL INFORMATION AND DECLARATIONS

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Competing Interests

The authors declare that they have no competing interests.

Author Contributions

- Thamolwan Poopradubsil conceived and designed the experiments, performed the experiments, analyzed the data, performed the computation work, prepared figures and/or tables, and approved the final draft.
- Sheng-Chih Chen conceived and designed the experiments, prepared figures and/or tables, and approved the final draft.
- Panon Latcharote conceived and designed the experiments, prepared figures and/or tables, and approved the final draft.
- Tipajin Thaipisutikul conceived and designed the experiments, performed the experiments, analyzed the data, performed the computation work, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.

Data Availability

The following information was supplied regarding data availability:

The Climate Policy Radar data is available at Zenodo: Poopradubsil, T., & Thaipisutikul, T. (2026). POLiMATCH Dataset [Data set]. Zenodo. https://doi.org/10.5281/zenodo.19915339.

The code is available at GitHub and Zenodo:

- https://github.com/thamolwanpo/SEEK-Policy.

- thamolwan.po, & Claude. (2026). thamolwanpo/SEEK-Policy: SEEK-Policy (v1.0.0). Zenodo. https://doi.org/10.5281/zenodo.20948341.

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